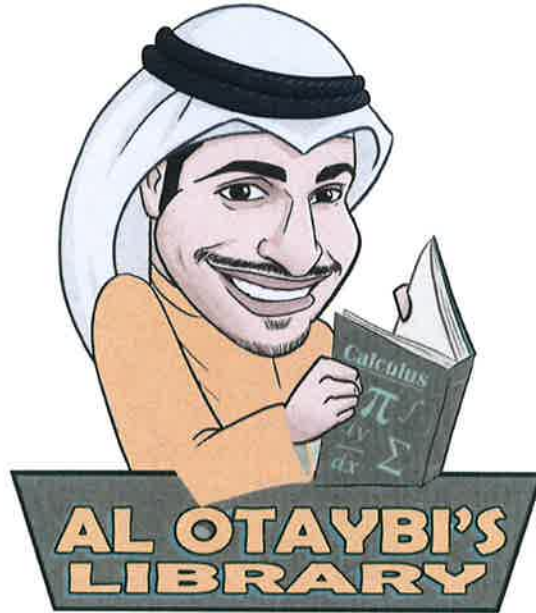


Kuwait University

2021/2022

Fall course



Calculus B

First

مراجعة

اختبارات سابقة

(مراجعة الفيرست)

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Spring 2019

Evaluate

$$\lim_{x \rightarrow 0} (\csc x - \cot x)$$

Find The partial decomposition of

$$\frac{x^2 + 3}{(x - 2)^2(x^2 + 2x + 4)}$$

Evaluate the integral.

$$\int \sec x \cdot \tan^3 x \, dx$$

Evaluate the integral.

$$\int \frac{\sin(8x)}{\sec(5x)} \, dx$$

Evaluate

$$\lim_{x \rightarrow 0} (2 - \cos(4x))^{(1/x^2)}$$

Evaluate the integral.

$$\int \sin(4x)e^x \, dx$$

Evaluate the integral.

$$\int \frac{t + 1}{t^2 \sqrt{t^2 - 16}} \, dt$$

Evaluate the integral.

$$\int \frac{\sin^2 x \cos x}{\sin^2 x + 5 \sin x - 6} \, dx$$

Evaluate the integral.

$$\int \frac{1}{3 \sin x + 4 \cos x} \, dx$$

Evaluate if possible ,

$$\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)} \, dx$$

Evaluate

$$\lim_{x \rightarrow 0} (\csc x - \cot x)$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right) \quad \frac{\frac{1}{0} - \frac{1}{0}}{\infty - \infty}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x} \quad \frac{1-1}{0} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{\sin x}{\cos x} = \frac{0}{1} = 0$$

$$\therefore \lim_{x \rightarrow 0} (\csc x - \cot x) = 0$$

Find The partial decomposition of

$$\frac{x^2 + 3}{(x - 2)^2(x^2 + 2x + 4)}$$

$$= \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{Cx + D}{x^2 + 2x + 4}$$

Evaluate the integral.

Spring 2019

$$\int \sec x \cdot \tan^3 x \, dx$$

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$$\int \tan^2 x \sec x \tan x \, dx$$

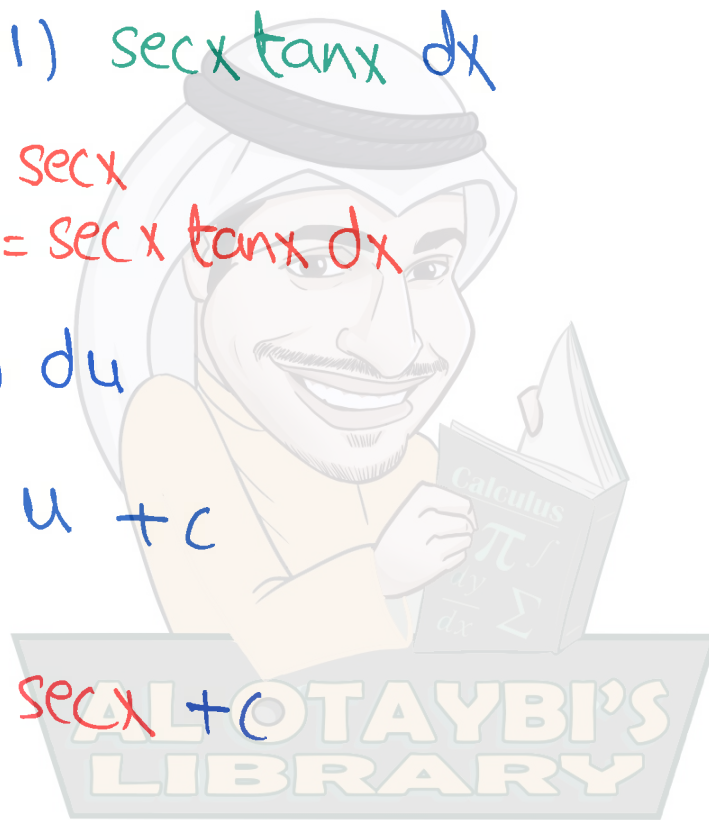
$$\int (\sec^2 x - 1) \sec x \tan x \, dx$$

$$\text{Let } u = \sec x \\ du = \sec x \tan x \, dx$$

$$\int (u^2 - 1) \, du$$

$$= \frac{u^3}{3} - u + C$$

$$= \frac{\sec^3 x}{3} - \sec x + C$$



Evaluate the integral.

$$\int \frac{\sin(8x)}{\sec(5x)} dx$$

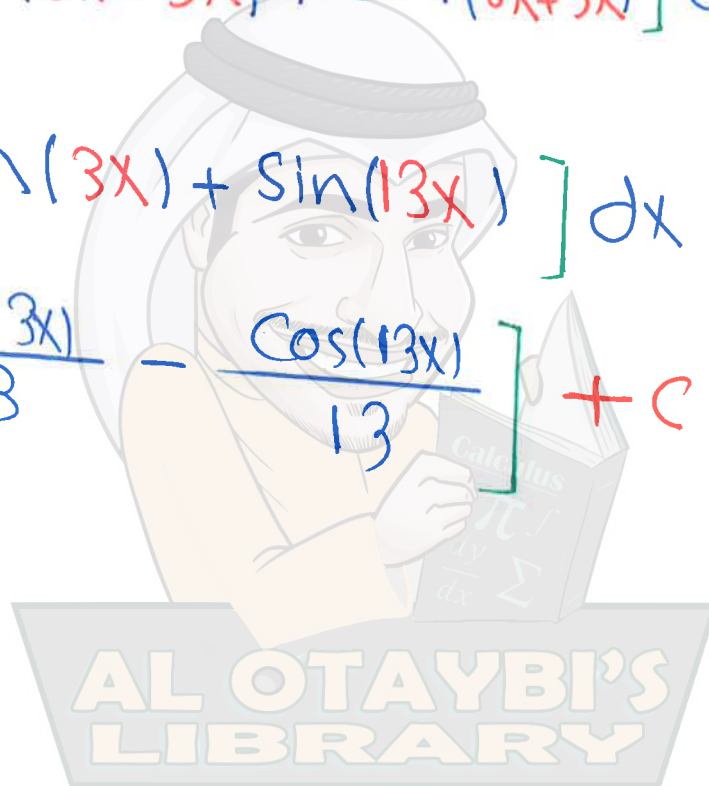
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$$\int \sin(8x) \cos(5x) dx$$

$$\frac{1}{2} \int [\sin(8x - 5x) + \sin(8x + 5x)] dx$$

$$\frac{1}{2} \int [\sin(3x) + \sin(13x)] dx$$

$$\frac{1}{2} \left[\frac{-\cos(3x)}{3} - \frac{\cos(13x)}{13} \right] + C$$



Evaluate

Spring 2019

$$\lim_{x \rightarrow 0} (2 - \cos(4x))^{(1/x^2)}$$

1^∞

$$y = (2 - \cos(4x))^{1/x^2}$$

$$\ln y = \ln (2 - \cos(4x))^{1/x^2}$$

$$\ln y = \frac{1}{x^2} \ln (2 - \cos(4x))$$

$$\lim_{x \rightarrow 0} \frac{\ln (2 - \cos(4x))}{x^2} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{\sin(4x) \cdot 4}{2 - \cos(4x)} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{4 \sin(4x)}{2x(2 - \cos(4x))} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{2 \cos(4x) \cdot 4}{(1)(2 - \cos(4x)) + x(\sin(4x) \cdot 4)}$$

$$= \frac{2(1)(4)}{(2-1) + 0} = \frac{8}{1} = 8$$

$$\therefore \lim_{x \rightarrow 0} (2 - \cos(4x))^{1/x^2} = e^8$$

Evaluate the integral.

$$\int \sin(4x) e^x dx = I$$

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$$u = \sin(4x) \quad dv = e^x dx$$

$$du = 4\cos(4x) dx \quad v = e^x$$

$$\sin(4x) e^x - 4 \int e^x \cos(4x) dx$$

$$u = \cos(4x) \quad dv = e^x dx$$

$$du = -4\sin(4x) dx \quad v = e^x$$

$$\cos(4x) e^x + 4 \int e^x \sin(4x) dx$$

$$I = \sin(4x) e^x - 4 \left[\cos(4x) e^x + 4I \right]$$

$$I = \sin(4x) e^x - 4 \cos(4x) e^x - 16I$$

$$17I = \sin(4x) e^x - 4 \cos(4x) e^x$$

$$I = \frac{1}{17} \left[\sin(4x) e^x - 4 \cos(4x) e^x \right] + C$$

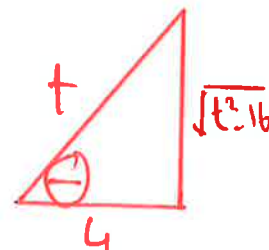
Evaluate the integral.

$$\int \frac{t+1}{t^2 \sqrt{t^2-16}} dt$$

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Let $t = 4 \sec \theta \longrightarrow \frac{t}{4} = \sec \theta$

$dt = 4 \sec \theta \tan \theta d\theta$



$\theta \in [0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2}]$

$$\int \frac{4 \sec \theta + 1}{(4 \sec \theta)^2 \sqrt{(4 \sec \theta)^2 - 16}} \cdot 4 \sec \theta \tan \theta d\theta$$

$$\int \frac{4 \sec \theta + 1}{16 \sec^2 \theta \sqrt{16 \sec^2 \theta - 16}} \cdot 4 \sec \theta \tan \theta d\theta$$

$\downarrow$
 $\sqrt{16(\sec^2 \theta - 1)} = \sqrt{16 \tan^2 \theta} = 4 \tan \theta$

$$\int \frac{4 \sec \theta + 1}{16 \sec^2 \theta} \cdot 4 \sec \theta \tan \theta d\theta$$

$$\int \frac{4 \sec \theta + 1}{16 \sec \theta} d\theta = \frac{1}{16} \int \left[\frac{4 \sec \theta}{\sec \theta} + \frac{1}{\sec \theta} \right] d\theta$$

$$\frac{1}{16} \int [4 + \cos \theta] d\theta$$

$$= [4\theta + \sin \theta] + C$$

$$= \left[4 \sec^{-1}\left(\frac{t}{4}\right) + \frac{\sqrt{t^2-16}}{t} \right] + C$$

Evaluate the integral.

Spring 2019

$$\int \frac{\sin^2 x \cos x}{\sin^2 x + 5 \sin x - 6} dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int \frac{u^2}{u^2 + 5u - 6} du$$
$$\int \left[1 + \frac{-5u+6}{u^2+5u-6} \right] du$$

$$\begin{array}{r} 1 \\ \hline u^2 \\ \ominus u^2 + 5u \oplus \\ \hline -5u + 6 \end{array}$$

$$\boxed{u} + \int \frac{-5u+6}{u^2+5u-6} du$$

$$\frac{-5u+6}{(u-1)(u+6)} = \frac{A}{u-1} + \frac{B}{u+6}$$

* $(u-1)(u+6)$

$$-5u+6 = A(u+6) + B(u-1)$$

$$\underline{u=1} \quad -5+6 = A(7) + 0$$

$$1 = 7A \longrightarrow A = \frac{1}{7}$$

$$\underline{u=-6} \quad 36 = 0 + B(-7) \longrightarrow B = \frac{-36}{7}$$

$$\int \left[\frac{\frac{1}{7}}{u-1} + \frac{\frac{-36}{7}}{u+6} \right] du$$

$$= \boxed{\frac{1}{7} \ln|u-1| - \frac{36}{7} \ln|u+6| + C}$$

$$\Rightarrow \sin x + \frac{1}{7} \ln|\sin x - 1| - \frac{36}{7} \ln|\sin x + 6| + C$$

Evaluate the integral.

Spring 2019

$$\int \frac{1}{3\sin x + 4\cos x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{3 \frac{2t}{1+t^2} + 4 \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int \frac{2}{6t + 4(1-t^2)} dt = \int \frac{2}{6t + 4 - 4t^2} dt$$

$$\int \frac{2}{-2(2t^2 - 3t - 2)} dt = - \int \frac{1}{2t^2 - 3t - 2} dt$$

$$\frac{1}{(2t+1)(t-2)} = \frac{A}{2t+1} + \frac{B}{t-2}$$

$$\times (2t+1)(t-2)$$

$$1 = A(t-2) + B(2t+1)$$

$$\underline{t=2} \quad 1 = 0 + B(5) \rightarrow B = \frac{1}{5}$$

$$\underline{t=-\frac{1}{2}} \quad 1 = A\left(-\frac{1}{2}-2\right) + 0 \rightarrow 1 = A\left(-\frac{5}{2}\right) \rightarrow A = -\frac{2}{5}$$

$$- \int \left[\frac{-\frac{2}{5}}{2t+1} + \frac{\frac{1}{5}}{t-2} \right] dt = - \int \left[\frac{\frac{2}{5}}{2t+1} + \frac{\frac{1}{5}}{t-2} \right] dt$$
$$= - \left[\frac{1}{5} \ln|2t+1| + \frac{1}{5} \ln|t-2| \right] + C$$

$$\Rightarrow - \left[\frac{1}{5} \ln|2\tan\left(\frac{x}{2}\right)+1| + \frac{1}{5} \ln|\tan\left(\frac{x}{2}\right)-2| \right] + C$$

Evaluate if possible ,

Spring 2019

$$\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)} dx$$

$$I_1 = \int \frac{1}{\sqrt{x}(1+x)} dx$$

0 ∞

$$y = \sqrt{x}$$

$$y^2 = x$$

$$2y dy = dx$$

$$\int \frac{1}{y(1+y^2)} 2y dy = 2 \tan^{-1} y + C$$

$$= 2 \tan^{-1} \sqrt{x} + C$$

$$\int_0^1 \frac{1}{\sqrt{x}(1+x)} dx + \int_1^{\infty} \frac{1}{\sqrt{x}(1+x)} dx$$

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{\sqrt{x}(1+x)} dx + \lim_{t \rightarrow \infty} \int_1^t \frac{1}{\sqrt{x}(1+x)} dx$$

$$\lim_{t \rightarrow 0^+} [2 \tan^{-1} \sqrt{x}]_t^1 + \lim_{t \rightarrow \infty} [2 \tan^{-1} \sqrt{x}]_1^t$$

$$2 \lim_{t \rightarrow 0^+} [\tan^{-1} 1 - \tan^{-1} \sqrt{t}] + 2 \lim_{t \rightarrow \infty} [\tan^{-1} \sqrt{t} - \tan^{-1} 1]$$

$$2 \left[\frac{\pi}{4} - 0 \right] + 2 \left[\frac{\pi}{2} - \frac{\pi}{4} \right] = \frac{\pi}{2} + \pi - \frac{\pi}{2} = \pi$$

I is convergent

Math 102: Calculus II

First Exam • 8 April 2017 • Duration: 90 Minutes

1. [5 pts each] Determine whether each statement is true or false, and justify your answer.

(a) $\frac{x - x^3}{(x - 1)(x^3 - 1)}$ can be put in the form $\frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{Cx + D}{x^2 + x + 1}$.

(b) $\int_0^2 (1 - x)^{-4/3} dx = -6$.

2. [15 pts] Evaluate $\lim_{x \rightarrow \infty} \left[\cos\left(\frac{1}{x}\right) \right]^{x^2}$.

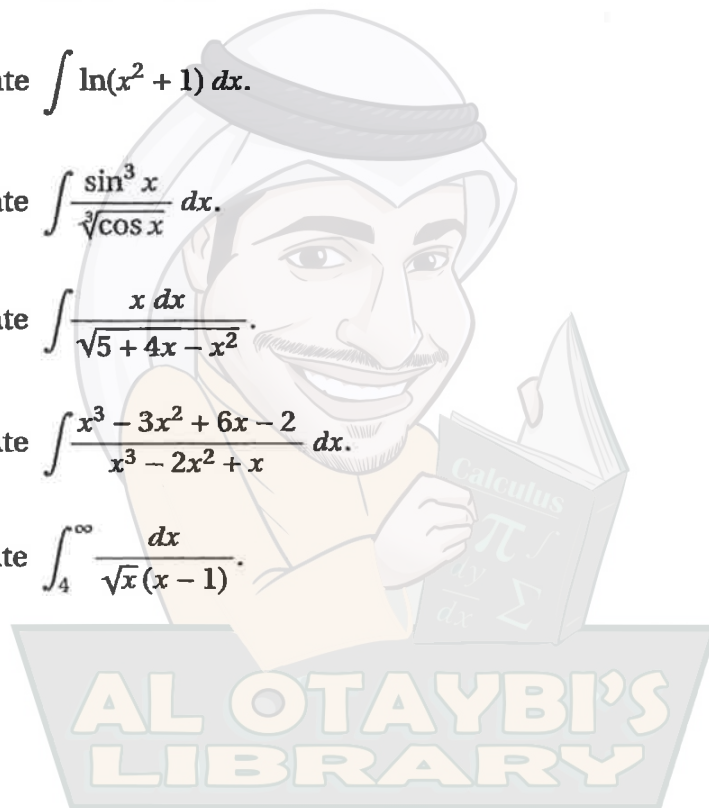
3. [15 pts] Evaluate $\int \ln(x^2 + 1) dx$.

4. [15 pts] Evaluate $\int \frac{\sin^3 x}{\sqrt{\cos x}} dx$.

5. [15 pts] Evaluate $\int \frac{x dx}{\sqrt{5 + 4x - x^2}}$.

6. [15 pts] Evaluate $\int \frac{x^3 - 3x^2 + 6x - 2}{x^3 - 2x^2 + x} dx$.

7. [15 pts] Evaluate $\int_4^{\infty} \frac{dx}{\sqrt{x}(x - 1)}$.



1)

$$a) \frac{x - x^3}{(x-1)(x^3-1)}$$

$$\frac{x - x^3}{(x-1)(x-1)(x^2+x+1)}$$

$$\frac{x - x^3}{(x-1)^2(x^2+x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+x+1}$$

True

$$b) \int_0^2 (1-x)^{-\frac{4}{3}} dx = -6$$

$$I_1 = \int (1-x)^{-\frac{4}{3}} dx = \frac{(1-x)^{-\frac{1}{3}}}{(-1) \cdot \frac{1}{3}} + C$$

$$= 3(1-x)^{-\frac{1}{3}} + C = \frac{3}{(1-x)^{\frac{1}{3}}} + C$$

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False

$$\int_0^1 (1-x)^{-\frac{4}{3}} dx + \int_1^2 (1-x)^{-\frac{4}{3}} dx$$

$$\lim_{t \rightarrow 1^-} \int_0^t (1-x)^{-\frac{4}{3}} dx + \lim_{t \rightarrow 1^+} \int_t^2 (1-x)^{-\frac{4}{3}} dx$$

$$\lim_{t \rightarrow 1^-} \left[\frac{3}{(1-x)^{\frac{1}{3}}} \right]_0^t + \lim_{t \rightarrow 1^+} \left[\frac{3}{(1-x)^{\frac{1}{3}}} \right]_t^2$$

$$\lim_{t \rightarrow 1^-} \left[\frac{3}{(1-t)^{\frac{1}{3}}} - \frac{3}{1} \right] + \lim_{t \rightarrow 1^+} \left[\frac{3}{(-1)} - \frac{3}{(1-t)^{\frac{1}{3}}} \right]$$

$$= \infty \rightarrow \infty \rightarrow \text{Div.}$$

$$2) \quad \lim_{x \rightarrow \infty} \left[\cos\left(\frac{1}{x}\right) \right]^{x^2} \quad 1^\infty$$

$$y = \left[\cos \frac{1}{x} \right]^{x^2}$$

$$\ln y = \ln \left[\cos \frac{1}{x} \right]^{x^2}$$

$$\ln y = x^2 \ln \left[\cos \frac{1}{x} \right]$$

$$\lim_{x \rightarrow \infty} x^2 \ln \cos \frac{1}{x}$$

$$\begin{array}{l} \infty \ln \cos 0 \\ \infty \ln 1 \\ \infty \cdot 0 \end{array}$$

$$\lim_{x \rightarrow \infty} \frac{\ln \cos \frac{1}{x}}{\frac{1}{x^2}} \quad \frac{\ln 1}{0} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow \infty} \frac{\ln \cos \frac{1}{x}}{x^{-2}} \quad \frac{-\sin \frac{1}{x}}{\cos \frac{1}{x}} \quad \frac{-\frac{1}{x^2}}{-2x^{-3}}$$

L'H

$$\lim_{x \rightarrow \infty} \frac{+\tan \frac{1}{x} \cdot \frac{1}{x^2}}{-\frac{2}{x^3}}$$

$$\lim_{x \rightarrow \infty} \frac{\tan \frac{1}{x} \cdot x^3}{-2x^2}$$

$$\frac{-1}{2} \quad \lim_{x \rightarrow \infty} x \tan \frac{1}{x}$$

$\infty \cdot 0$

$$\frac{-1}{2} \quad \lim_{x \rightarrow \infty} \frac{\tan \frac{1}{x}}{\frac{1}{x}} \quad \frac{0}{0}$$

L, H

$$\frac{-1}{2} \quad \lim_{x \rightarrow \infty} \frac{\sec^2 \frac{1}{x}}{\frac{1}{x^2}} \quad \left(\frac{1}{x^2} \right)$$

$\frac{1}{x^2}$

$$\frac{-1}{2} \quad \sec^2 0$$

$$\frac{-1}{2}$$

$$(1)^2$$

$$=$$

$$\frac{-1}{2}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left[\cos \frac{1}{x} \right]^{x^2} = e^{-\frac{1}{2}}$$

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3)

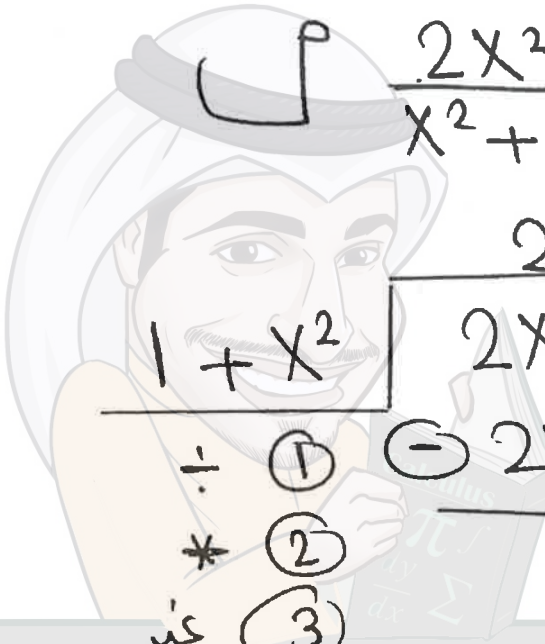
$$\int \ln(x^2+1) dx$$

$$u = \ln(x^2+1) \quad \swarrow \quad dv = dx$$

$$du = \frac{2x}{x^2+1} dx \quad \nwarrow \quad v = x$$

$$\ln(x^2+1) x - \int x \frac{2x}{x^2+1} dx$$

$$\int \frac{2x^2}{x^2+1} dx$$



1 + x ²	2x ²
÷ ①	- 2x ² + 2
* ②	
غير ③	- 2
+ ④	

$$\int \left[2 + \frac{-2}{x^2+1} \right] dx$$

$$2x - 2 \tan^{-1} x + c$$

$$\Rightarrow \ln(x^2+1) x - [2x - 2 \tan^{-1} x] + c$$

4)

$$\int \frac{\sin^3 x}{\sqrt[3]{\cos x}} dx$$

$$\int \sin^3 x (\cos x)^{-\frac{1}{3}} dx$$

$$\int \sin^2 x (\cos x)^{-\frac{1}{3}} \sin x dx$$

$$\int (1 - \cos^2 x) (\cos x)^{-\frac{1}{3}} \sin x dx$$

$$u = \cos x$$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$-\int (1 - u^2) u^{-\frac{1}{3}} du$$

$$-\int (u^{-\frac{1}{3}} - u^{\frac{5}{3}}) du$$

$$-\left[\frac{u^{\frac{2}{3}}}{\frac{2}{3}} - \frac{u^{\frac{8}{3}}}{\frac{8}{3}} \right] + C$$

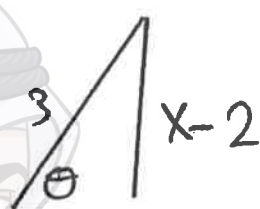
$$-\left[\frac{3}{2} (\cos x)^{\frac{2}{3}} - \frac{3}{8} (\cos x)^{\frac{8}{3}} \right] + C$$

$$5) \int \frac{x}{\sqrt{5+4x-x^2}} dx$$

$$\int \frac{x}{\sqrt{9-(x-2)^2}} dx \quad \begin{matrix} \swarrow \\ -[x^2-4x-5] \\ \searrow \\ -[(x-2)^2-4-5] \\ \text{E.P.} \end{matrix}$$

Let $x-2 = 3 \sin \theta \rightarrow \frac{x-2}{3} = \sin \theta \quad [(x-2)^2 - 9]$
 $dx = 3 \cos \theta d\theta$
 $9 - (x-2)^2$

$x = 3 \sin \theta + 2$



$$\int \frac{3 \sin \theta + 2}{\sqrt{9 - (3 \sin \theta)^2}} \cdot 3 \cos \theta d\theta$$

$$\begin{aligned} &\rightarrow \sqrt{9 - 9 \sin^2 \theta} = \sqrt{9(1 - \sin^2 \theta)} \\ &\sqrt{9 \cos^2 \theta} = 3 \cos \theta \end{aligned}$$

$$\int \frac{3 \sin \theta + 2}{\cancel{3 \cos \theta}} \cancel{3 \cos \theta} d\theta$$

$$= 3 \cos \theta + 2\theta + C$$

$$= 3 \frac{\sqrt{9-(x-2)^2}}{3} + 2 \sin^{-1} \left(\frac{x-2}{3} \right) + C$$

6)

$$\int \frac{x^3 - 3x^2 + 6x - 2}{x^3 - 2x^2 + x} dx$$

$$\begin{array}{r} 1 \\ x^3 - 2x^2 + x \overline{) x^3 - 3x^2 + 6x - 2} \\ \underline{-(x^3 - 2x^2 + x)} \\ -x^2 + 5x - 2 \end{array}$$

$$\int \left[1 + \frac{-x^2 + 5x - 2}{x^3 - 2x^2 + x} \right] dx$$

$$x + \int \frac{-x^2 + 5x - 2}{x(x^2 - 2x + 1)} dx$$

$(x-1)(x-1)$

$$\frac{-x^2 + 5x - 2}{x(x-1)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$* x(x-1)^2$$

$$-x^2 + 5x - 2 = A(x-1)^2 + Bx(x-1) + Cx$$

$$x=0 \quad -2 = A(-1)^2 + 0 + 0 \rightarrow A = -2$$

$$x=1 \quad -1 + 5 - 2 = 0 + 0 + C \rightarrow 2 = C$$

Equate coeff. of x^2

$$-1 = A + B$$

$$-1 = -2 + B \rightarrow B = 1$$

$$\int \left[\frac{-2}{x} + \frac{1}{x-1} + \frac{2}{(x-1)^2} \right] dx$$

$$-2 \ln|x| + \ln|x-1| + \frac{2(x-1)^{-1}}{-1} + C$$



$$x + (-2 \ln|x| + \ln|x-1| - 2(x-1)^{-1}) + C$$



$$7) \int_4^{\infty} \frac{dx}{\sqrt{x}(x-1)}$$

∞

$$I_1 = \int \frac{dx}{\sqrt{x}(x-1)}$$

$$y = \sqrt{x}$$

$$y^2 = x$$

$$2y dy = dx$$

$$I_1 = \int \frac{2y}{y(y^2-1)} dy = \frac{1}{2} \ln \left| \frac{y-1}{y+1} \right| + C$$

$$= \ln \left| \frac{\sqrt{x}-1}{\sqrt{x}+1} \right| + C$$

$$\lim_{t \rightarrow \infty} \int_4^t \frac{dx}{\sqrt{x}(x-1)}$$

$$\lim_{t \rightarrow \infty} \left[\ln \left| \frac{\sqrt{x}-1}{\sqrt{x}+1} \right| \right]_4^t$$

$$\lim_{t \rightarrow \infty} \left[\ln \left| \frac{\sqrt{t}-1}{\sqrt{t}+1} \right| - \ln \frac{1}{3} \right]$$

L.Hop

$$\lim_{t \rightarrow \infty} \ln \left(\frac{\frac{1}{\sqrt{t}}}{\frac{1}{\sqrt{t}}} \right)$$

$$= \ln(1)$$

$$= 0$$

$$\Rightarrow \therefore I = \left[0 - \ln \frac{1}{3} \right]$$

$$= - \ln \frac{1}{3}$$

$\therefore I$ is convergent



Math 102: Calculus II

Mid-Term Exam I: October 15, 2018

Duration: 90 minutes

- In question 1, select the correct answer (no justification is required).
- In questions 2-7, show all your work.
- Calculators and communication devices are not allowed in the examination room.

1. [$5 \times 5 = 25$ pts.] In each of the following multiple choice questions, select the correct answer.

I. $\int \sin(5x) \cos(3x) dx =$

(A) $\frac{\cos(5x) \sin(3x)}{15} + C$

(B) $-\frac{\cos(5x) \sin(3x)}{15} + C$

(C) $\frac{\cos(8x)}{16} - \frac{\cos(2x)}{4} + C$

(D) $-\frac{\cos(8x)}{16} - \frac{\cos(2x)}{4} + C$

(E) None of the above

II. The partial partial decomposition of $\frac{x^4 + 3x^2 + 2}{x^2(x^2 - x + 1)^2}$ is

(A) $\frac{A}{x} + \frac{Bx + C}{x^2 - x + 1}$

(B) $\frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 - x + 1}$

(C) $\frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 - x + 1} + \frac{Ex + F}{(x^2 - x + 1)^2}$

(D) $\frac{A}{x^2} + \frac{Bx + C}{x^2 - x + 1} + \frac{Dx + E}{(x^2 - x + 1)^2}$

(E) None of the above

III. Let a and b be two real numbers such that $a \neq 0$. Then $\int x e^{ax+b} dx =$

(A) $\frac{x^2 e^{ax+b}}{2a} + C$

(B) $\frac{e^{ax+b}}{a} \left(x - \frac{1}{a} \right) + C$

(C) $\frac{e^{ax+b}}{a} \left(x + \frac{1}{a} \right) + C$

(D) $\frac{e^{ax+b}}{a} \left(\frac{1}{a} - x \right) + C$

(E) None of the above

IV. $\lim_{x \rightarrow 1^-} \left[(x-1) \tan \left(\frac{\pi x}{2} \right) \right] =$

(A) $\pi/2$

(B) π

(C) $-2/\pi$

(D) $-\pi$

(E) None of these

V. $\int_0^{\pi/3} \sec^5 x \tan x dx =$

(A) $31/5$

(B) $-31/5$

(C) $5/31$

(D) $-5/31$

(E) None of these

2. [10 pts.] Evaluate $\lim_{x \rightarrow +\infty} (3e^x + 2x^2)^{4/x}$.

The limit is indeterminate of the form ∞^0 . Let $y = (3e^x + 2x^2)^{4/x}$ and hence $\ln y = \frac{4}{x} \ln (3e^x + 2x^2)$.

Then, we obtain: $\lim_{x \rightarrow +\infty} \ln y = \lim_{x \rightarrow +\infty} \frac{4 \ln (3e^x + 2x^2)}{x}$ which is indeterminate of the form ∞/∞ . Invoking l'Hospital's rule, we have: $\lim_{x \rightarrow +\infty} \ln y = \lim_{x \rightarrow +\infty} \frac{4(3e^x + 4x)}{3e^x + 2x^2} = 4$. Therefore $\lim_{x \rightarrow +\infty} y = e^4$.

3. [10 pts.] Find $\int \sin(\ln(x^3)) dx$.

Let I be the given integral. Then, integrating by parts with $u = \sin(\ln(x^3)) = \sin(3 \ln x)$ and $dv = dx$, we obtain: $I = x \sin(3 \ln x) - 3 \int \cos(3 \ln x) dx$. Another integration by parts gives: $I = x \sin(3 \ln x) - 3 [x \cos(3 \ln x) + 3I]$. Thereby $10I = x \sin(3 \ln x) - 3x \cos(3 \ln x) + C$ and hence $\int \sin(\ln(x^3)) dx = \frac{1}{10} [x \sin(3 \ln x) - 3x \cos(3 \ln x)] + C$.

4. [10 pts.] Find $\int \frac{\sqrt{x}}{1+x^3} dx$.

Let $u = \sqrt{x}$. This implies that $u^2 = x$ and $2u du = dx$. Thus $\int \frac{\sqrt{x}}{1+x^3} dx = 2 \int \frac{u^2}{1+u^6} du = \frac{2}{3} \int \frac{dv}{1+v^2} dv = \frac{2}{3} \tan^{-1} v + C$, where $v = u^3$. Hence $\int \frac{\sqrt{x}}{1+x^3} dx = \frac{2}{3} \tan^{-1}(x^{3/2}) + C$.

(N.B: One simply may use the substitution $u = x^{3/2}$).

5. [15 pts.] Evaluate $\int \frac{x}{\sqrt{x^2 - 2x - 8}} dx$.

We have: $\int \frac{x}{\sqrt{x^2 - 2x - 8}} dx = \int \frac{x}{\sqrt{(x-1)^2 - 9}} dx = \int \frac{(3 \sec \theta + 1)(3 \sec \theta \tan \theta)}{3 \tan \theta} d\theta$, where $x - 1 = 3 \sec \theta$.

The last integral gives: $\int (3 \sec^2 \theta + \sec \theta) d\theta = 3 \tan \theta + \ln |\sec \theta + \tan \theta| + C$. Finally, we have:

$$\int \frac{x}{\sqrt{x^2 - 2x - 8}} dx = \sqrt{x^2 - 2x - 8} + \ln |x - 1 + \sqrt{x^2 - 2x - 8}| + K, \text{ where } K = C - \ln 3.$$

6. [15 pts.] Evaluate $\int \frac{e^{4x}}{2e^x + e^{2x} - 3} dx$.

Let $u = e^x$. The integral writes: $\int \frac{u^3}{u^2 + 2u - 3} du$. Using long division, we obtain:

$$\int \frac{u^3}{u^2 + 2u - 3} du = \int (u - 2) du + \int \frac{7u - 6}{(u - 1)(u + 3)} du. \text{ Then, partial fractions gives:}$$

$$\frac{7u - 6}{(u - 1)(u + 3)} = \frac{7/4}{u - 1} + \frac{27/4}{u + 3}. \text{ Therefore,}$$

$$\int \frac{u^3}{u^2 + 2u - 3} du = \frac{u^2}{2} - 2u + \frac{1}{4} \ln |u - 1| + \frac{27}{4} \ln |u + 3| + C.$$

$$\text{Finally, we obtain: } \int \frac{e^{4x}}{e^{2x} + 3e^x + 2} dx = \frac{e^{2x}}{2} - 2e^x + \frac{1}{4} \ln |e^x - 1| + \frac{27}{4} \ln(e^x + 1) + C.$$

7. [15 pts.] Evaluate $\int_0^{+\infty} \frac{dx}{\sqrt{x}e^{\sqrt{x}}}$ if possible.

As the interval is $[0, \infty)$ and the integrand has an infinite discontinuity at $x = 0$, the integral is improper and can be written as follows:

$$\int_1^{\infty} \frac{1}{\sqrt{x}e^{\sqrt{x}}} dx = \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{\sqrt{x}e^{\sqrt{x}}} dx + \lim_{t \rightarrow \infty} \int_1^t \frac{1}{\sqrt{x}e^{\sqrt{x}}} dx.$$

$$\text{It turns out that } \int \frac{1}{\sqrt{x}e^{\sqrt{x}}} dx \stackrel{(u=-\sqrt{x})}{=} 2 \int e^{-u} du = -2e^{-\sqrt{x}} + C.$$

$$\text{Whereupon, } \int_1^{\infty} \frac{1}{\sqrt{x}e^{\sqrt{x}}} dx = -2 \left[\lim_{a \rightarrow 0^+} (e^{-1} - e^{-\sqrt{a}}) + \lim_{t \rightarrow +\infty} (e^{-\sqrt{t}} - e^{-1}) \right] = 2.$$

Therefore the improper integral converges.

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① $\int \sin(5x) \cos(3x) dx$

$$\frac{1}{2} \int [\sin(5x-3x) + \sin(5x+3x)] dx$$

$$\frac{1}{2} \int [\sin(2x) + \sin(8x)] dx$$

$$\frac{1}{2} \left[-\frac{\cos(2x)}{2} - \frac{\cos(8x)}{8} \right] + C$$

$$= -\frac{\cos(2x)}{4} - \frac{\cos(8x)}{16} + C$$

$\Rightarrow (D) \checkmark$

②

The partial decomposition of

$$\frac{x^4 + 3x^2 + 2}{x^2(x^2 - x + 1)^2}$$

$$= \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 - x + 1} + \frac{Ex + F}{(x^2 - x + 1)^2}$$

$\Rightarrow (C) \checkmark$

$$(III) \int x e^{ax+b} dx$$

$$u = x$$

$$dv = e^{ax+b} dx$$

$$du = dx$$

$$v = \frac{e^{ax+b}}{a}$$

$$\frac{1}{a} x e^{ax+b} - \frac{1}{a} \int e^{ax+b} dx$$

$$\frac{1}{a} x e^{ax+b} - \frac{1}{a} \frac{e^{ax+b}}{a} + C$$

$$\frac{1}{a} e^{ax+b} \left[x - \frac{1}{a} \right] + C$$

$\Rightarrow (B) \checkmark$

$$V) \int_0^{\frac{\pi}{3}} \sec^5 x \tan x dx$$

$$\int_0^{\frac{\pi}{3}} \sec^4 x \overset{\text{زیر}}{\sec x} \overset{\text{فری}}{\tan x} dx$$

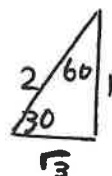
$$u = \sec x$$

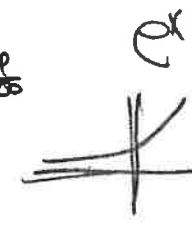
$$du = \sec x \tan x dx$$

$$\int u^4 du = \frac{u^5}{5} = \frac{1}{5} \left[(\sec x)^5 \right]_0^{\frac{\pi}{3}}$$

$$\frac{1}{5} \left[\left(\frac{2}{1} \right)^5 - (1)^5 \right] = \frac{1}{5} [32 - 1]$$

$$= \frac{31}{5} \Rightarrow (A)$$



(2) $\lim_{x \rightarrow \infty} (3e^x + 2x^2)^{\frac{4}{x}} \quad (\infty + \infty)^{\frac{4}{\infty}}$ $\frac{\infty}{\infty}$ 

$$y = (3e^x + 2x^2)^{\frac{4}{x}}$$

$$\ln y = \ln (3e^x + 2x^2)^{\frac{4}{x}}$$

$$\ln y = \frac{4}{x} \ln (3e^x + 2x^2)$$

$\lim_{x \rightarrow \infty} \frac{4 \ln (3e^x + 2x^2)}{x} \quad \frac{4 \ln \infty}{\infty} \quad \frac{\infty}{\infty}$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4 \frac{3e^x + 4x}{3e^x + 2x^2}}{1}$$

$$\lim_{x \rightarrow \infty} \frac{4 (3e^x + 4x)}{3e^x + 2x^2} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4 (3e^x + 4)}{3e^x + 4x} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4 (3e^x)}{3e^x + 4}$$

أفتر أهله عوافل e^x

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4 (3e^x)}{3e^x} = \frac{4(3)}{3} = 4$$

$$\lim_{x \rightarrow \infty} (3e^x + 2x^2)^{\frac{4}{x}} = e^4$$

(3) $\int \sin(\ln(x^3)) dx$ ————— أسهل السؤال
بتحليل القواعد

$$\int \sin(3 \ln x) dx = I$$

$u = \sin(3 \ln x)$ $dv = dx$ لولا

$du = \cos(3 \ln x) \cdot \frac{3}{x} dx$ $\swarrow \searrow$ $v = x$

$$\sin(3 \ln x) x - 3 \int \cancel{x} \cos(3 \ln x) \frac{1}{x} dx$$

$u = \cos(3 \ln x)$ $dv = dx$

$du = -\sin(3 \ln x) \cdot \frac{3}{x} dx$ $\swarrow \searrow$ $v = x$

$\cos(3 \ln x) \cdot x + 3 \int \cancel{x} \sin(3 \ln x) \frac{1}{x} dx$
I

$$I = \sin(3 \ln x) x - 3 [\cos(3 \ln x) x + 3 I]$$

$$I = \sin(3 \ln x) x - 3 \cos(3 \ln x) x - 9 I$$

$$10 I = \sin(3 \ln x) x - 3 \cos(3 \ln x) x$$

$$I = \frac{1}{10} [\sin(3 \ln x) x - 3 \cos(3 \ln x) x] + C$$

$$(4) \int \frac{\sqrt{x}}{1+x^3} dx$$

$$\text{Let } y = \sqrt{x}$$

$$y^2 = x$$

$$2y dy = dx$$

$$\begin{aligned} (y^2)^3 &= x^3 \\ y^6 &= x^3 \end{aligned}$$

$$\int \frac{y}{1+y^6} 2y dy$$

$$2 \int \frac{y^2}{1+y^6} dy$$

$$2 \int \frac{y^2}{1+(y^3)^2} dy$$

$$\text{Let } u = y^3$$

$$du = 3y^2 dy$$

$$2 \cdot \frac{1}{3} \int \frac{1}{1+u^2} du \rightarrow \frac{2}{3} \int \frac{1}{1+u^2} du = \frac{2}{3} \tan^{-1} u$$

$$= \frac{2}{3} \tan^{-1} u + C$$

$$= \frac{2}{3} \tan^{-1} y^3 + C$$

$$= \frac{2}{3} \tan^{-1} (\sqrt{x})^3 + C$$

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$$(4) \int \frac{x}{\sqrt{x^2 - 2x - 8}} dx$$

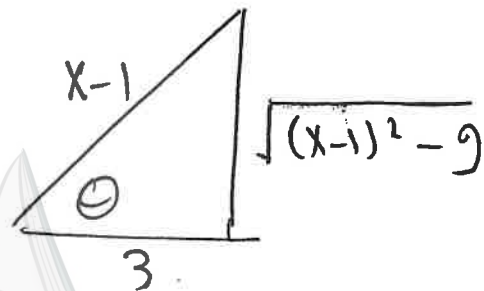
$$\int \frac{x}{\sqrt{(x-1)^2 - 9}} dx \quad \xrightarrow{\quad} \quad \begin{array}{l} x^2 - 2x - 8 \\ \downarrow \\ (x-1)^2 - 1 - 8 \\ \downarrow \\ (x-1)^2 - 9 \end{array}$$

Let $x-1 = 3 \sec \theta$

$$dx = 3 \sec \theta \tan \theta d\theta \quad \frac{x-1}{3} = \sec \theta$$

$$x = 3 \sec \theta + 1$$

$$\int \frac{3 \sec \theta + 1}{\sqrt{(3 \sec \theta)^2 - 9}} \cdot 3 \sec \theta \tan \theta d\theta$$



$$\begin{aligned} \sqrt{9 \sec^2 \theta - 9} &= \sqrt{9(\sec^2 \theta - 1)} = \sqrt{9 \tan^2 \theta} \\ &= 3 \tan \theta \end{aligned}$$

$$\int \frac{3 \sec \theta + 1}{3 \tan \theta} \cdot 3 \sec \theta \tan \theta d\theta$$

$$\int (3 \sec^2 \theta + \sec \theta) d\theta$$

$$3 \tan \theta + \ln |\sec \theta + \tan \theta| + C$$

$$3 \frac{\sqrt{(x-1)^2 - 9}}{3} + \ln \left| \frac{x-1}{3} + \frac{\sqrt{(x-1)^2 - 9}}{3} \right| + C$$

$$(6) \int \frac{e^{4x}}{2e^x + e^{2x} - 3} dx$$

Let $u = e^x$
 $du = e^x dx$

$$\int \frac{e^{3x} \cdot \cancel{e^x} \cdot \cancel{dx}}{2e^x + e^{2x} - 3}$$

$$\int \frac{u^3}{2u + u^2 - 3} du$$

$$\int \frac{u^3}{u^2 + 2u - 3} du$$

Partial Fraction

$$\int \left[u - 2 + \frac{7u - 6}{u^2 + 2u - 3} \right] du$$

$$\frac{u^2}{2} - 2u + \int \frac{\frac{7}{2}(2u+2) - 13}{u^2 + 2u - 3} du$$

$$\int \left(\frac{\frac{7}{2}(2u+2)}{u^2 + 2u - 3} + \frac{-13}{u^2 + 2u - 3} \right) du$$

$$\frac{7}{2} \ln|u^2 + 2u - 3|$$

$$\int \frac{-13}{(u+1)^2 - 4} du$$

$$\frac{u^2}{2} - 2u + \frac{7}{2} \ln|u^2 + 2u - 3| - 13 \ln \left| \frac{u+1-2}{u+1+2} \right| + C$$

$$= \frac{e^{2x}}{2} - 2e^x + \frac{7}{2} \ln|e^{2x} + 2e^x - 3| - 13 \ln \left| \frac{e^x - 1}{e^x + 3} \right| + C$$

$$(7) \int_0^{\infty} \frac{dx}{\sqrt{x} e^{\sqrt{x}}}$$

⑦ ⑧

$$I_1 = \int \frac{dx}{\sqrt{x} e^{\sqrt{x}}}$$

Let $u = \sqrt{x}$
 $du = \frac{1}{2\sqrt{x}} dx$
 $2du = \frac{1}{\sqrt{x}} dx$

$$2 \int \frac{1}{e^u} du$$

$$2 \int e^{-u} du = 2 \frac{e^{-u}}{-1} + C$$

$$-2 e^{-\sqrt{x}} + C$$

$$\int_0^1 \frac{dx}{\sqrt{x} e^{\sqrt{x}}} + \int_1^{\infty} \frac{dx}{\sqrt{x} e^{\sqrt{x}}}$$

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{dx}{\sqrt{x} e^{\sqrt{x}}} + \lim_{t \rightarrow \infty} \int_1^t \frac{dx}{\sqrt{x} e^{\sqrt{x}}}$$

$$\lim_{t \rightarrow 0^+} [-2e^{-\sqrt{x}}]_t^1 + \lim_{t \rightarrow \infty} [-2e^{-\sqrt{x}}]_1^t$$

$$-2 \lim_{t \rightarrow 0^+} [e^{-1} - e^{-\sqrt{t}}] - 2 \lim_{t \rightarrow \infty} [e^{-\sqrt{t}} - e^{-1}]$$

$$-2 [e^{-1} - e^0] - 2 [e^{-\infty} - e^{-1}]$$

$$-2 [e^{-1} - 1] - 2 [0 - e^{-1}]$$

$$-2e^{-1} + 2 + 2e^{-1} = 2 \Rightarrow \therefore \text{Convergent}$$



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 102: Calculus II

Fall Semester 2017-18

First Midterm, 30 October 2017

Duration: 90 minutes

- In questions 1-10, circle the correct answer. No justification is required.
- In questions 11-13, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1-10 (6 points each)	
Q11 (10 points)	
Q12 (15 points)	
Q13 (15 points)	
Total	

1. $\lim_{x \rightarrow 0} \frac{x2^x}{2^x - 1}$

- a) does not exist.
- b) equals to $-\ln 2$.
- c) equals to $1/\ln 2$.
- d) equals to $2\ln 2$.
- e) equals to $\ln 2$.

2. $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$ is

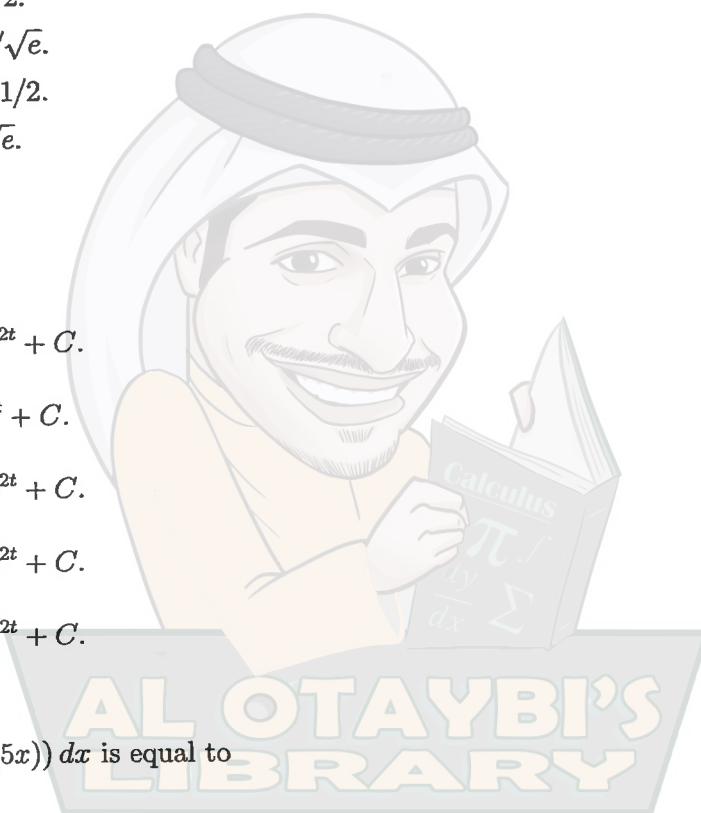
- a) equal to $-\sqrt{e}$.
- b) equal to $1/2$.
- c) equal to $1/\sqrt{e}$.
- d) equal to $-1/2$.
- e) equal to $\sqrt{e}$.

3. $\int te^{2t} dt$ equals

- a) $\frac{1}{4}(2t + 2)e^{2t} + C$.
- b) $\frac{1}{4}(t + 1)e^{2t} + C$.
- c) $\frac{1}{4}(2t + 1)e^{2t} + C$.
- d) $\frac{1}{2}(2t + 1)e^{2t} + C$.
- e) $\frac{1}{4}(2t - 1)e^{2t} + C$.

4. $\int (\sin(3x))(\cos(5x)) dx$ is equal to

- a) $-\frac{\sin(8x)}{16} - \frac{\cos(2x)}{4} + C$.
- b) $\frac{\cos(8x)}{16} + \frac{\cos(2x)}{4} + C$.
- c) $-\frac{\cos(8x)}{16} + \frac{\sin(2x)}{4} + C$.
- d) $-\frac{\cos(8x)}{16} + \frac{\cos(2x)}{4} + C$.
- e) $\frac{\sin(8x)}{16} - \frac{\cos(2x)}{2} + C$.



5. The partial fraction decomposition of $\frac{x^2 - 6}{x^3 + 8}$ is

- a) $\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$.
- b) $\frac{A}{x+2} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$.
- c) $\frac{A}{x+2} + \frac{Bx+C}{x^2-2x+4}$.
- d) $\frac{A}{x-2} + \frac{Bx+C}{x^2+2x+4}$.
- e) $\frac{A}{x+2} + \frac{Bx+C}{x^2+2x+4}$.

6. $\int \frac{1}{x + \sqrt[3]{x}} dx$ is equal to

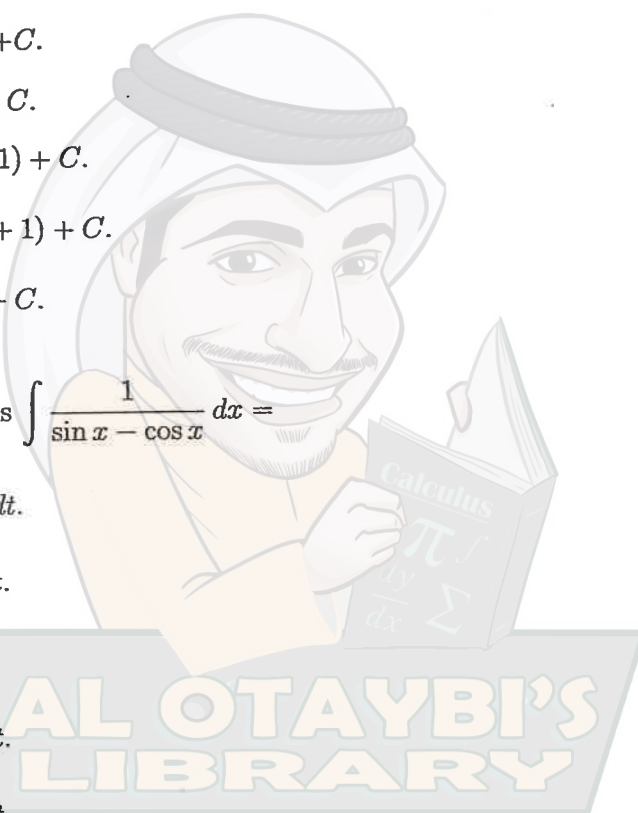
- a) $\frac{3}{2} \ln |\sqrt[3]{x} + 1| + C$.
- b) $3 \ln(\sqrt[3]{x^2} + 1) + C$.
- c) $\frac{3}{2} \tan^{-1}(\sqrt[3]{x^2} + 1) + C$.
- d) $-\frac{3}{2} \tan^{-1}(\sqrt[3]{x^2} + 1) + C$.
- e) $\frac{3}{2} \ln(\sqrt[3]{x^2} + 1) + C$.

7. Using $t = \tan \frac{x}{2}$ gives $\int \frac{1}{\sin x - \cos x} dx =$

- a) $\int \frac{1}{2t^2 + 2t - 1} dt$.
- b) $\int \frac{1}{t^2 - 2t - 1} dt$.
- c) $\int \frac{2}{(t+1)^2} dt$.
- d) $\int \frac{2}{t^2 - 2t + 1} dt$.
- e) $\int \frac{2}{t^2 + 2t - 1} dt$.

8. $\int_{-\infty}^3 e^{x/3} dx$

- a) converges to e^3 .
- b) converges to $9e$.
- c) converges to $e/3$.
- d) diverges.
- e) converges to $3e$.



$$1) \lim_{x \rightarrow 0} \frac{x 2^x}{2^x - 1} \quad \frac{0}{1-1} \quad \frac{0}{0}$$

L.H

$$\lim_{x \rightarrow 0} \frac{\text{اشتق (1)} \quad \text{اشتق} \quad \text{اشتق} \quad \text{اشتق}}{2^x \ln 2} = \frac{1 + 0}{2^x \ln 2}$$

$$= \frac{1 + 0}{(1) \ln 2} = \frac{1}{\ln 2}$$

(C)

$$2) \lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}} \quad 1^{\frac{1}{0}} \quad 1^{\infty}$$

$$y = (\cos x)^{\frac{1}{x^2}}$$

$$\ln y = \ln (\cos x)^{\frac{1}{x^2}}$$

$$\ln y = \frac{1}{x^2} \ln \cos x$$

$$\lim_{x \rightarrow 0} \frac{\ln \cos x}{x^2} \quad \frac{\ln 1}{0} \quad \frac{0}{0}$$

L.H

$$\lim_{x \rightarrow 0} \frac{-\sin x}{2x}$$

$$\lim_{x \rightarrow 0} \frac{-\tan x}{2x}$$

L.H

$$\lim_{x \rightarrow 0} \frac{-\sec^2 x}{2} = -\frac{(1)^2}{2} = -\frac{1}{2}$$

$$\boxed{e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}}$$

0/0

(C)

3)

$$\int t e^{2t} dt$$

$$\begin{array}{l} u = t \quad \swarrow \quad \searrow \quad \begin{array}{l} dv = e^{2t} dt \\ v = \frac{e^{2t}}{2} \end{array} \\ du = dt \quad \swarrow \quad \searrow \end{array}$$

$$\frac{1}{2} t e^{2t} - \frac{1}{2} \int e^{2t} dt$$

$$\frac{1}{2} t e^{2t} - \frac{1}{2} \frac{e^{2t}}{2} + C$$

$$\frac{1}{2} t e^{2t} - \frac{1}{4} e^{2t} + C$$

$$\frac{1}{4} [2t e^{2t} - e^{2t}] + C \quad \textcircled{c}$$

$$\int \sin(3x) \cos(5x) dx$$

$$\frac{1}{2} \int [\sin(3x-5x) - \sin(3x+5x)] dx$$

$$\frac{1}{2} \int [\sin(-2x) - \sin(8x)] dx$$

$$\frac{1}{2} \int [-\sin(2x) - \sin(8x)] dx$$

$$\frac{1}{2} \left[\frac{\cos 2x}{2} - \frac{\cos 8x}{8} \right] + C$$

$$\frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C \quad \textcircled{d}$$

51

$$\frac{x^2 - 6}{x^3 + 8}$$

$$\frac{x^2 - 6}{(x+2)(x^2 - 2x + 4)} = \frac{A}{x+2} + \frac{Bx + C}{x^2 - 2x + 4}$$

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$$6) \int \frac{1}{x + \sqrt[3]{x}} dx$$

$$y = \sqrt[3]{x}$$

$$y^3 = x$$

$$3y^2 dy = dx$$

$$\int \frac{1}{y^3 + y} \cdot 3y^2 dy$$

$$\int \frac{1}{y(y^2 + 1)} 3y^2 dy$$

$$\int \frac{3y}{y^2 + 1} dy$$

$$\int \frac{\frac{3}{2} (2y)}{y^2 + 1} dy$$

$$\frac{3}{2} \ln |y^2 + 1| + C$$

$$\frac{3}{2} \ln |(\sqrt[3]{x})^2 + 1| + C$$

©

7)

$$\int \frac{1}{\sin x - \cos x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{\frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int \frac{2}{2t - (1-t^2)} dt$$

$$\downarrow$$

$$2t - 1 + t^2$$

$$\int \frac{2}{t^2 + 2t - 1} dt \quad (e)$$

8)

$$\int_{-\infty}^3 e^{\frac{x}{3}} dx$$

$$I_1 = \int e^{\frac{x}{3}} dx = \frac{e^{\frac{x}{3}}}{\frac{1}{3}} + c = 3e^{\frac{x}{3}} + c$$

$$\lim_{t \rightarrow -\infty} \int_t^3 e^{\frac{x}{3}} dx$$

$$\lim_{t \rightarrow -\infty} [3e^{\frac{x}{3}}]_t^3 = 3 \lim_{t \rightarrow -\infty} [e^1 - e^{\frac{t}{3}}]$$

$$= 3 [e^1 - \cancel{e^{-\infty}} \rightarrow 0]$$

$$= 3e \quad (e)$$

$$g) \int_1^2 \frac{1}{\sqrt{(x-1)^5}} dx$$

الحرج
①

$$I_1 = \int \frac{1}{(x-1)^{\frac{5}{2}}} dx = \int (x-1)^{-\frac{5}{2}} dx$$

$$= \frac{(x-1)^{-\frac{3}{2}}}{-\frac{3}{2}} + C = -\frac{2}{3} \cdot \frac{1}{(x-1)^{\frac{3}{2}}} + C$$

$$\lim_{t \rightarrow 1^+} \int_t^2 \frac{1}{\sqrt{(x-1)^5}} dx$$

$$\lim_{t \rightarrow 1^+} \left[-\frac{2}{3} \cdot \frac{1}{(x-1)^{\frac{3}{2}}} \right]_t^2$$

$$\frac{-2}{3} \lim_{t \rightarrow 1^+} \left[\frac{1}{1} - \frac{1}{(t-1)^{\frac{3}{2}}} \right]$$

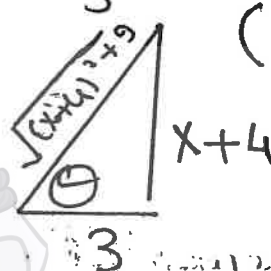
$$-\frac{2}{3} \left[1 - \frac{1}{0} \right]$$

$$-\frac{2}{3} [1 - \infty] = \infty \Rightarrow \underline{\underline{\text{Div.}}}$$

ⓐ

$$12) \int \frac{(x+4)^3}{\sqrt{x^2+8x+25}} dx$$

$$\int \frac{(x+4)^3}{\sqrt{(x+4)^2+9}} dx \quad \begin{matrix} \searrow \\ x^2+8x+25 \\ \downarrow \\ (x+4)^2-16+25 \end{matrix}$$

Let $x+4 = 3 \tan \theta \rightarrow \frac{x+4}{3} = \tan \theta$ 

$$dx = 3 \sec^2 \theta d\theta$$

$$\int \frac{(3 \tan \theta)^3}{\sqrt{(3 \tan \theta)^2+9}} \cdot 3 \sec^2 \theta d\theta$$

$$\begin{matrix} \searrow \\ \sqrt{9 \tan^2 \theta + 9} = \sqrt{9(\tan^2 \theta + 1)} = \sqrt{9 \sec^2 \theta} \\ = 3 \sec \theta \end{matrix}$$

$$\int \frac{27 \tan^3 \theta}{\cancel{3 \sec \theta}} \cdot \cancel{3 \sec^2 \theta} d\theta$$

$$27 \int \tan^3 \theta \sec \theta d\theta$$

فردی $\swarrow$ $\searrow$ ~~زوجی~~

$$27 \int \tan^2 \theta \sec \theta \tan \theta d\theta$$

$$27 \int (\sec^2 \theta - 1) \sec \theta \tan \theta d\theta$$

$$u = \sec \theta$$

$$du = \sec \theta \tan \theta d\theta$$

$$27 \int (u^2 - 1) du = 27 \left[\frac{u^3}{3} - u \right] + c$$

$$27 \left[\frac{(\sec \theta)^3}{3} - \sec \theta \right] + c = 27 \left[\frac{(\sqrt{(x+4)^2+9})^3}{3} - \frac{\sqrt{(x+4)^2+9}}{3} \right] + c$$

$$13) \int \ln(x^2 + 2x + 10) dx$$

$$u = \ln(x^2 + 2x + 10)$$

$$dv = dx$$

$$du = \frac{2x + 2}{x^2 + 2x + 10} dx \quad v = x$$

$$\ln(x^2 + 2x + 10) \cdot x - \int x \cdot \frac{2x + 2}{x^2 + 2x + 10} dx$$

$$\int \frac{2x^2 + 2x}{x^2 + 2x + 10} dx$$

$$\begin{array}{r} 2 \\ 10 + 2x + x^2 \overline{) 2x^2 + 2x} \\ \underline{- 2x^2 \quad - 4x \quad - 20} \\ -2x - 20 \end{array}$$

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* ②
غير ③
+ ④

$$\int \left[2 + \frac{-2x - 20}{x^2 + 2x + 10} \right] dx$$

$$2x + \int \frac{-2x - 20}{x^2 + 2x + 10} dx$$

← see next page
Plz!

$$\int \frac{-2x - 20}{x^2 + 2x + 10} dx$$

$$\int \frac{-(2x + 2) - 18}{x^2 + 2x + 10} dx$$

$$\int \left[\frac{-(2x + 2)}{x^2 + 2x + 10} + \frac{-18}{x^2 + 2x + 10} \right] dx$$

$$- \ln|x^2 + 2x + 10| + \int \frac{-18}{x^2 + 2x + 10} dx$$

$\downarrow x^{\frac{1}{2}}$
 $(x + 1)^2 - 1 + 10$
 عر -

$$\int \frac{-18}{(x + 1)^2 + 9} dx$$

$$-18 \cdot \frac{1}{3} \tan^{-1}\left(\frac{x+1}{3}\right) + C$$

$$\ln(x^2 + 2x + 10) x - \left[2x - \ln|x^2 + 2x + 10| - 6 \tan^{-1}\left(\frac{x+1}{3}\right) \right] + C$$

Math 102: Calculus II

First Exam • 5 November 2016 • 9–10:30 am

*Calculators and communication devices are not allowed in the examination room.
Answer all questions and justify all your answers.*

Q1 [15 pts] Find $\lim_{x \rightarrow 0^+} \frac{\arcsin \sqrt{x}}{1 - \ln(e - x)}$.

Evaluate the following integrals.

Q2 [15 pts] $\int x^2 4^x dx$

Q3 [15 pts] $\int \frac{x dx}{\sqrt{x^2 - 4x + 3}}$

Q4 [15 pts] $\int \frac{dx}{2 \sin x - \cos x}$

Q5 [15 pts] $\int \sin^5 x \cos^3 x dx$

Q6 [15 pts] $\int_e^\infty \frac{dx}{x (\ln x)^3}$

Q7 [10 pts] $\int_0^\pi \sqrt{1 + \sin x} dx$



1)

$$\lim_{x \rightarrow 0^+} \frac{\sin^{-1} \sqrt{x}}{1 - \ln(e-x)}$$

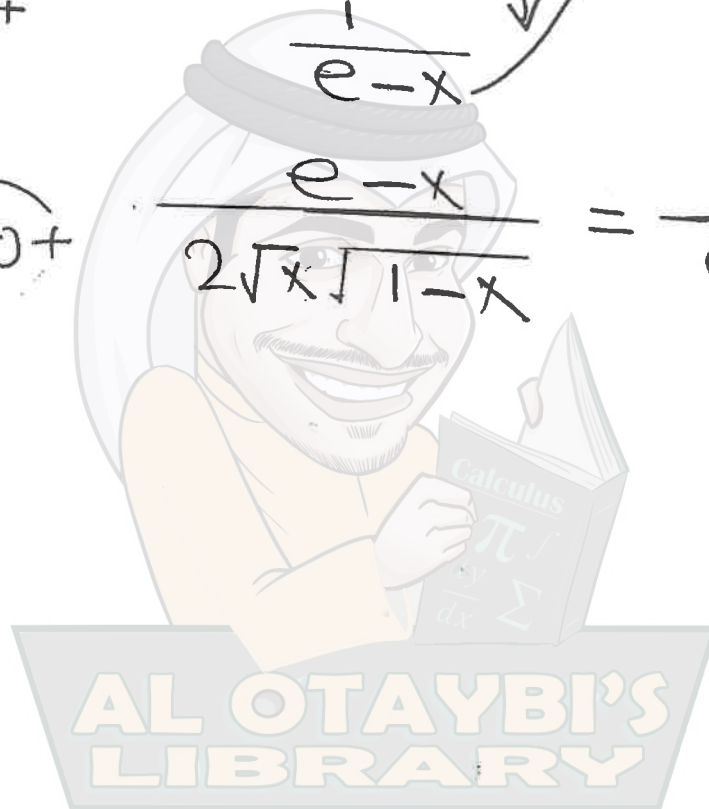
$$\frac{\sin^{-1} 0}{1 - \ln e} \quad \frac{0}{1-1} \quad \frac{0}{0}$$

L.H

$$\lim_{x \rightarrow 0^+} \frac{\frac{\frac{1}{2\sqrt{x}}}{\sqrt{1-(\sqrt{x})^2}}}{- \frac{-1}{e-x}}$$

$$\lim_{x \rightarrow 0^+} \frac{\frac{1}{2\sqrt{x}\sqrt{1-x}}}{\frac{1}{e-x}}$$

$$\lim_{x \rightarrow 0^+} \frac{e-x}{2\sqrt{x}\sqrt{1-x}} = \frac{e}{0} = \infty$$



$$2) \int x^2 4^x dx$$

$$u = x^2 \quad \begin{array}{l} \rightarrow \\ \leftarrow \end{array} \quad \begin{array}{l} dv = 4^x dx \\ v = \frac{4^x}{\ln 4} \end{array}$$

$$du = 2x dx \quad \leftarrow$$

$$\frac{1}{\ln 4} x^2 4^x - \frac{2}{\ln 4} \int 4^x x dx$$

$$\int 4^x x dx$$

$$u = x \quad \begin{array}{l} \rightarrow \\ \leftarrow \end{array} \quad \begin{array}{l} dv = 4^x dx \\ v = \frac{4^x}{\ln 4} \end{array}$$

$$du = dx \quad \leftarrow$$

$$\frac{1}{\ln 4} x \cdot 4^x - \frac{1}{\ln 4} \int 4^x dx$$

$$\frac{1}{\ln 4} x 4^x - \frac{1}{\ln 4} \frac{4^x}{\ln 4} + C$$

$$\Rightarrow \frac{1}{\ln 4} x^2 4^x - \frac{2}{\ln 4} \left[\frac{1}{\ln 4} x 4^x - \frac{1}{\ln 4} \frac{4^x}{\ln 4} \right] + C$$

3)

$$\int \frac{x}{\sqrt{x^2 - 4x + 3}} dx$$

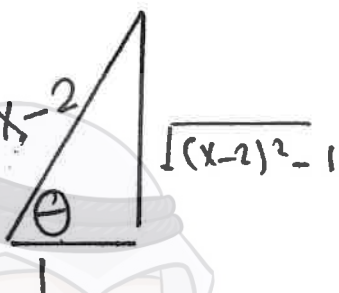
$$\int \frac{x}{\sqrt{(x-2)^2 - 1}} dx \quad \xrightarrow{\quad} \quad \begin{array}{l} x^2 - 4x + 3 \\ \downarrow \times \frac{1}{2} \\ (x-2)^2 - 4 + 3 \\ \text{p.p.} \quad \hookrightarrow \\ (x-2)^2 - 1 \end{array}$$

Let

$$x-2 = \sec \theta$$

$$dx = \sec \theta \tan \theta d\theta$$

$$x = \sec \theta + 2$$



$$\int \frac{\sec \theta + 2}{\sqrt{\sec^2 \theta - 1}} \cdot \sec \theta \tan \theta d\theta$$

$$\xrightarrow{\quad} \quad \sqrt{\tan^2 \theta} = \tan \theta$$

$$\int \frac{\sec \theta + 2}{\cancel{\tan \theta}} \sec \theta \cancel{\tan \theta} d\theta$$

$$\int (\sec^2 \theta + 2 \sec \theta) d\theta$$

$$\tan \theta + 2 \ln |\sec \theta + \tan \theta| + C$$

$$\frac{\sqrt{(x-2)^2 - 1}}{1} + 2 \ln \left| \frac{x-2}{1} + \frac{\sqrt{(x-2)^2 - 1}}{1} \right| + C$$

4)

$$\int \frac{dx}{2 \sin x - \cos x}$$

$$\int \frac{1}{2 \sin x - \cos x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int \frac{1}{2 \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int \frac{2}{2 \cdot 2t - (1-t^2)} dt$$

$$\int \frac{2}{4t - 1 + t^2} dt$$

$$\int \frac{2}{t^2 + 4t - 1} dt$$

$$(t+2)^2 - 4 - 1$$

$$\int \frac{2}{(t+2)^2 - 5} dt$$

$$2 \cdot \frac{1}{2\sqrt{5}} \ln \left| \frac{t+2-\sqrt{5}}{t+2+\sqrt{5}} \right| + C$$

$$\frac{1}{\sqrt{5}} \ln \left| \frac{\tan \frac{x}{2} + 2 - \sqrt{5}}{\tan \frac{x}{2} + 2 + \sqrt{5}} \right| + C$$

$$6) \int_e^{\infty} \frac{dx}{x (\ln x)^3}$$

$$I_1 = \int \frac{dx}{x (\ln x)^3}$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{u^3} du = \int u^{-3} du = \frac{u^{-2}}{-2} + C = -\frac{1}{2} u^{-2} + C$$

$$= -\frac{1}{2} \frac{1}{(\ln x)^2} + C$$

$$\lim_{t \rightarrow \infty} \int_e^t \frac{dx}{x (\ln x)^3}$$

$$\lim_{t \rightarrow \infty} \left[-\frac{1}{2} \frac{1}{(\ln x)^2} \right]_e^t$$

$$= -\frac{1}{2} \lim_{t \rightarrow \infty} \left[\frac{1}{(\ln t)^2} - \frac{1}{(1)^2} \right]$$

$$= -\frac{1}{2} \left[\frac{1}{\infty} - 1 \right] = -\frac{1}{2} (-1) = \left(\frac{1}{2} \right)$$

Convergent

∞

$$x (\ln x)^3 = 0$$

$$x=0 \quad \ln x=0$$

$$x=0$$

$$x=1$$

$\neq$

$\neq$

Math 102: Calculus II

First Exam • 5 March 2016 • 9–10:30 am

*Calculators and communication devices are not allowed in the examination room.
Answer all questions and justify all your answers.*

1. [10 pts] Find $\lim_{x \rightarrow -\infty} (x^2 - x^3) e^{2x}$.

2. [10 pts] Write out the form of the partial fraction decomposition of the function

$$\frac{x + 9 - x^9}{(x - 1)(x + 1)^2(x^2 + 1)^3(x^2 - 2x + 7)}.$$

(Do not compute the coefficients.)

3. [10 pts] Evaluate $\int x^3 \sin(x^2) dx$.

4. [10 pts] Evaluate $\int \frac{\sec^4 x}{\sqrt[5]{\tan x}} dx$.

5. [15 pts] Evaluate $\int \frac{3x + 2}{\sqrt{x^2 - 2x - 8}} dx$.

6. [15 pts] Evaluate $\int_0^{\pi/2} \frac{dx}{3 + \cos x - 2 \sin x}$.

7. [15 pts] Use a trigonometric substitution to evaluate $\int \frac{2}{(x^2 + 1)^2} dx$.

8. [15 pts] Determine whether $\int_0^{\ln 2} \frac{dx}{\sqrt{e^x - 1}}$ is convergent or divergent.

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1)

$$\lim_{x \rightarrow -\infty} (x^2 - x^3) e^{2x} \quad (\infty + \infty) e^{-\infty}$$

$\infty \quad 0$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - x^3}{\frac{1}{e^{2x}}} \quad \frac{\infty + \infty}{\frac{1}{0}} \quad \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow -\infty} \frac{x^2 - x^3}{e^{-2x}}$$

L, H

$$\lim_{x \rightarrow -\infty} \frac{2x - 3x^2}{-2e^{-2x}} \quad \frac{-\infty - \infty}{-2e^{\infty}} \quad \frac{-\infty}{-\infty}$$

L, H

$$\lim_{x \rightarrow -\infty} \frac{2 - 6x}{4e^{-2x}} \quad \frac{\infty}{e^{\infty}} \quad \frac{\infty}{\infty}$$

L, H

$$\lim_{x \rightarrow -\infty} \frac{-6}{-8e^{-2x}} = \frac{-6}{-8e^{\infty}} = \frac{-6}{-\infty}$$

$= 0$

∴

$$\lim_{x \rightarrow -\infty} (x^2 - x^3) e^{2x} = 0$$

2.)

$$\frac{x + 9 - x^2}{(x-1)(x+1)^2(x^2+1)^3(x^2-2x+7)}$$

$$= \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{Dx+E}{x^2+1} + \frac{Fx+J}{(x^2+1)^2} + \frac{Hx+I}{(x^2+1)^3} + \frac{Gx+K}{x^2-2x+7}$$

3) $\int x^3 \sin(x^2) dx$

Let $t = x^2$

$$dt = 2x dx \rightarrow \frac{1}{2} dt = x dx$$

$\int x \cdot x^2 \sin(x^2) dx$

$$\frac{1}{2} \int t \sin t dt$$

$$\begin{array}{l} u=t \quad \rightarrow \quad dv = \sin t dt \\ du=dt \quad \leftarrow \quad v = -\cos t \end{array}$$

$$-t \cos t + \int \cos t dt$$

$$-t \cos t + \sin t + c$$

$$\frac{1}{2} [-t \cos t + \sin t] + c$$

$$\frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + c$$

$$4) \int \frac{\sec^4 x}{\sqrt[5]{\tan x}} dx$$

$$\int \frac{\sec^4 x}{\sqrt[5]{\tan x}} dx$$

← زوی
→ قوی

$$\int \sec^2 x (\tan x)^{-\frac{1}{5}} \sec^2 x dx$$

$$\int (1 + \tan^2 x) (\tan x)^{-\frac{1}{5}} \sec^2 x dx$$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$\int (1 + u^2) u^{-\frac{1}{5}} du$$

$$\int (u^{-\frac{1}{5}} + u^{\frac{9}{5}}) du$$

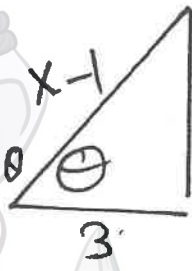
$$\frac{5}{4} u^{\frac{4}{5}} + \frac{5}{14} u^{\frac{14}{5}} + C$$

$$\frac{5}{4} (\tan x)^{\frac{4}{5}} + \frac{5}{14} (\tan x)^{\frac{14}{5}} + C$$

$$5) \int \frac{3x+2}{\sqrt{x^2-2x-8}} dx$$

$$\int \frac{3x+2}{\sqrt{(x-1)^2-9}} dx \quad \xrightarrow{\quad} \quad \begin{aligned} &x^2-2x-8 \\ &\quad \downarrow +1 \\ &(x-1)^2-1-8 \\ &\quad \quad \quad \downarrow \\ &\quad \quad \quad (x-1)^2-9 \end{aligned}$$

Let $x-1 = 3 \sec \theta \rightarrow$
 $dx = 3 \sec \theta \tan \theta d\theta$ $\frac{x-1}{3} = \sec \theta$
 $x = 3 \sec \theta + 1$

$$\int \frac{3(3 \sec \theta + 1) + 2}{\sqrt{(3 \sec \theta)^2 - 9}} \cdot 3 \sec \theta \tan \theta d\theta$$


$$\xrightarrow{\quad} \begin{aligned} &\sqrt{9 \sec^2 \theta - 9} = \sqrt{9(\sec^2 \theta - 1)} \\ &\sqrt{9 \tan^2 \theta} = 3 \tan \theta \end{aligned}$$

$$\int \frac{9 \sec \theta + 3 + 2}{\cancel{3 \tan \theta}} \cdot \cancel{3 \sec \theta \tan \theta} d\theta$$

$$\int (9 \sec \theta + 5) \sec \theta d\theta$$

$$\int (9 \sec^2 \theta + 5 \sec \theta) d\theta$$

$$9 \tan \theta + 5 \ln |\sec \theta + \tan \theta| + C$$

$$\int \frac{\sqrt{(x-1)^2-9}}{3} + 5 \ln \left| \frac{x-1}{3} + \frac{\sqrt{(x-1)^2-9}}{3} \right| + C$$

$$6) \int_0^{\frac{\pi}{2}} \frac{1}{3 + \cos x - 2 \sin x} dx$$

Let $t = \tan\left(\frac{x}{2}\right)$

$$\int_0^{\frac{\pi}{2}} \frac{1}{3 + \frac{1-t^2}{1+t^2} - 2 \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$\int_0^{\frac{\pi}{2}} \frac{2}{3(1+t^2) + 1 - t^2 - 4t} dt$$

$$3 + 3t^2 + 1 - t^2 - 4t$$

$$2t^2 - 4t + 4$$

$$2(t^2 - 2t + 2)$$

$$\int \frac{2}{2(t^2 - 2t + 2)} dt$$

$$\int \frac{1}{(t-1)^2 + 1} dt$$

$$= \tan^{-1}(t-1)$$

$$= \left(\tan^{-1}\left(\tan\left(\frac{x}{2}\right) - 1\right) \right)_0^{\frac{\pi}{2}}$$

$$= \tan^{-1}(1-1) - \tan^{-1}(-1)$$

$$= 0 + \pi_a = \frac{\pi}{4}$$

$$8) \int_0^{\ln 2} \frac{dx}{\sqrt{e^x - 1}}$$

$$\begin{aligned} e^x - 1 &= 0 \\ e^x &= 1 \\ \ln e^x &= \ln 1 \\ x &= 0 \in [0, \ln 2] \end{aligned}$$

Sol #1

$$I_1 = \int \frac{dx}{\sqrt{e^x - 1}}$$

مسائل

$$y = \sqrt{e^x - 1}$$

$$y^2 = e^x - 1$$

$$2y dy = e^x dx \rightarrow e^x = y^2 + 1$$

$$\int \frac{\frac{e^x}{e^x} \cdot dx}{\sqrt{e^x - 1}}$$

$$2 \int \frac{y}{(y^2 + 1) y} dy = 2 \tan^{-1} y + C = 2 \tan^{-1} \sqrt{e^x - 1} + C$$

$$\lim_{t \rightarrow 0^+} \int_t^{\ln 2} \frac{dx}{\sqrt{e^x - 1}}$$

$$\lim_{t \rightarrow 0^+} [2 \tan^{-1} \sqrt{e^x - 1}]_t^{\ln 2}$$

$$2 \lim_{t \rightarrow 0^+} [\tan^{-1} \sqrt{e^{\ln 2} - 1} - \tan^{-1} \sqrt{e^t - 1}]$$

$$2 [\tan^{-1} 1 - \tan^{-1} 0]$$

$$2 \left[\frac{\pi}{4} - 0 \right] = \left(\frac{\pi}{2} \right) \rightarrow \text{Convergent}$$

$$\int_0^{\ln 2} \frac{dx}{\sqrt{e^x - 1}}$$

Sol #2

$$I_1 = \int \frac{dx}{\sqrt{(e^{\frac{1}{2}x})^2 - 1}}$$

Let $u = e^{\frac{1}{2}x}$

$$du = \frac{1}{2} e^{\frac{1}{2}x} dx \rightarrow 2du = e^{\frac{1}{2}x} dx$$

$$\int \frac{e^{\frac{1}{2}x}}{e^{\frac{1}{2}x} \sqrt{(e^{\frac{1}{2}x})^2 - 1}} dx = 2 \int \frac{1}{u \sqrt{u^2 - 1}} du$$

$$= 2 \sec^{-1} u + C = 2 \sec^{-1} e^{\frac{1}{2}x} + C$$

$$\lim_{t \rightarrow 0^+} \int_t^{\ln 2} \frac{dx}{\sqrt{e^x - 1}}$$

$$\lim_{t \rightarrow 0^+} \left[2 \sec^{-1} e^{\frac{1}{2}x} \right]_t^{\ln 2}$$

$$2 \lim_{t \rightarrow 0^+} \left[\sec^{-1} e^{\frac{1}{2} \ln 2} - \sec^{-1} e^{\frac{1}{2}t} \right]$$

$$2 \left[\sec^{-1} e^{\ln 2^{\frac{1}{2}}} - \sec^{-1}(1) \right]$$

$$2 \left[\sec^{-1} 2^{\frac{1}{2}} - 0 \right]$$

$$2 \left[\sec^{-1} \sqrt{2} \right] = 2 \frac{\pi}{4} = \frac{\pi}{2}$$

Converges

KUWAIT UNIVERSITY
College of Science
Department of Mathematics

Calculus II (Math 102)

First in-term test

25 October 2015

Duration 90 minutes

Answer all questions and justify your answers.

Total points: 100.

Calculators and communication devices are not allowed.

1. (10 points) Determine whether the following limit exists. If the limit exists, evaluate it.

$$\lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{1}{\ln x} \right)$$

2. (15 points) Evaluate $\int \frac{x + 6\sqrt{x} + 9}{x^2\sqrt{x} + 4x\sqrt{x}} dx$.

3. (15 points) Find $\int \sqrt{\sec^5 x} \tan^3 x dx$.

4. (15 points) Evaluate $\int x\sqrt{5+4x-x^2} dx$.

5. (15 points) Calculate $\int_0^{\pi/4} (\cos 3x) \sin^2 x dx$.

6. (15 points) Find $\int (\cos 2x) \ln(\tan x) dx$.

7. (15 points) Determine whether the following integral is convergent or divergent. If the integral is convergent, evaluate it.

$$\int_e^\infty \frac{dx}{x(\ln x)^5}$$

1)

$$\lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{1}{\ln x} \right) \quad \frac{\frac{1}{0} - \frac{1}{0}}{\infty - \infty}$$

$$\lim_{x \rightarrow 1} \left(\frac{\ln x}{(x-1) \ln x} - \frac{(x-1)}{(x-1) \ln x} \right)$$

$$\lim_{x \rightarrow 1} \frac{\ln x - (x-1)}{(x-1) \ln x} \quad \frac{0}{0}$$

L.H $\lim_{x \rightarrow 1} \frac{\frac{1}{x} - 1}{(1) \ln x + (x-1) \frac{1}{x}}$

الشيء بشت + بشت اشتقا

فليق

$$\lim_{x \rightarrow 1} \frac{1-x}{x \ln x + x - 1}$$

$$\lim_{x \rightarrow 1} \frac{1-x}{x \ln x + x - 1} \quad \frac{0}{0}$$

L.H $\lim_{x \rightarrow 1} \frac{-1}{(1) \ln x + x \frac{1}{x} + 1}$

$$= \frac{-1}{0 + 1 + 1} = \frac{-1}{2}$$

$$\therefore \lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{1}{\ln x} \right) = \frac{-1}{2}$$

2)

$$\int \frac{x + 6\sqrt{x} + 9}{x^2 \sqrt{x} + 4x\sqrt{x}} dx$$

$$y = \sqrt{x}$$

$$y^2 = x$$

$$2y dy = dx$$

$$\int \frac{y^2 + 6y + 9}{(y^2)^2 y + 4y^2 y} \cdot 2y dy$$

$$2 \int \frac{y^2 + 6y + 9}{y^5 + 4y^3} y dy$$

$$2 \int \frac{y^2 + 6y + 9}{y^3 [y^2 + 4]} y dy$$

$$2 \int \frac{y^2 + 6y + 9}{y^2 [y^2 + 4]} dy$$

$$\frac{y^2 + 6y + 9}{y^2 [y^2 + 4]} = \frac{A}{y} + \frac{B}{y^2} + \frac{Cy + D}{y^2 + 4}$$

$$\times y^2 [y^2 + 4]$$

HW

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$$y^2 + 6y + 9 = A y [y^2 + 4] + B [y^2 + 4] + (Cy + D) y^2$$

$y=0$ $0+0+9 = 0 + B(4) + 0 \rightarrow B = \frac{9}{4}$

Equate the coeff. of y^3

$$0 = A + C \rightarrow (1)$$

ڪاٺ

Equate the coeff. of y^2

$$1 = B + D$$

$$1 = \frac{9}{4} + D \rightarrow D = -\frac{5}{4}$$

ڪاٺ

Equate the coeff. of y

$$6 = 4A \rightarrow A = \frac{6}{4} \rightarrow A = \frac{3}{2}$$

in (1) $0 = \frac{3}{2} + C$

$$C = -\frac{3}{2}$$

$$\int \left[\frac{\frac{3}{2}}{y} + \frac{\frac{9}{4}}{y^2} + \frac{-\frac{3}{2}y - \frac{5}{4}}{y^2 + 4} \right] dy$$

$$\frac{3}{2} \ln|y| + \frac{9}{4} \frac{y^{-1}}{-1} + \int \frac{-\frac{3}{2}(2y) - \frac{5}{4}}{y^2 + 4} dy$$

$$\int \left(\frac{-\frac{3}{2}(2y)}{y^2 + 4} + \frac{-\frac{5}{4}}{y^2 + 4} \right) dy$$

$$-\frac{3}{4} \ln|y^2 + 4| - \frac{5}{4} \frac{1}{2} \tan^{-1}\left(\frac{y}{2}\right) + C$$

$\sqrt{x}$

$$3) \int \sqrt{\sec^5 x} \tan^3 x \, dx$$

$$\int \sec^{\frac{5}{2}} x \tan^3 x \, dx$$

$\swarrow$ زوجی $\searrow$ فردی

$$\int \sec^{\frac{5}{2}-1} x \tan^2 x \sec x \tan x \, dx$$

$$\int \sec^{\frac{3}{2}} x (\sec^2 x - 1) \sec x \tan x \, dx$$

$$u = \sec x$$

$$du = \sec x \tan x \, dx$$

$$\int u^{\frac{3}{2}} (u^2 - 1) \, du$$

$$\int (u^{\frac{7}{2}} - u^{\frac{3}{2}}) \, du$$

$$\frac{2}{9} u^{\frac{9}{2}} - \frac{2}{5} u^{\frac{5}{2}} + C$$

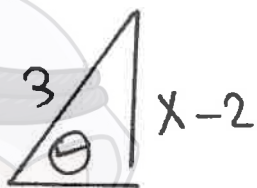
$$\frac{2}{9} (\sec x)^{\frac{9}{2}} - \frac{2}{5} (\sec x)^{\frac{5}{2}} + C$$

4) $\int x \sqrt{5 + 4x - x^2} \, dx$

$$\int x \sqrt{9 - (x-2)^2} dx \quad \xrightarrow{\quad} \quad -[x^2 - 4x - 5]$$

Let $x - 2 = 3 \sin \theta \rightarrow \frac{x-2}{3} = \sin \theta$
 $dx = 3 \cos \theta d\theta$

$\downarrow$
 $x = 3 \sin \theta + 2$



$9 - (x-2)^2$

$\int (3 \sin \theta + 2) \frac{\sqrt{9 - (3 \sin \theta)^2}}{\sqrt{9 - (3 \sin \theta)^2}} \cdot 3 \cos \theta d\theta$

$\hookrightarrow \sqrt{9 - 9 \sin^2 \theta} = \sqrt{9(1 - \sin^2 \theta)}$

$$\sqrt{9 \cos^2 \theta} = 3 \cos \theta$$

$$\int (3\sin\theta + 2) 3\cos\theta \cdot 3\cos\theta \, d\theta$$

$$\int 27 \sin \theta \cos^2 \theta d\theta + \int 18 \cos^2 \theta d\theta$$

$$u = \cos \theta$$

$$du = -\sin \theta \, d\theta$$

$$-du = \sin \theta d\theta$$

$$-27 \int u^2 dv = -27 \frac{u^3}{3}$$

$$-9 \cos^3 \theta + C_1$$

$$\frac{-9 \cos^3 \theta}{\sqrt{9 - (x-2)^2}} + C_1 \rightarrow -9 \left(\frac{\sqrt{9 - (x-2)^2}}{3} \right)^3 + 9 \left[\sin^{-1} \left(\frac{x-2}{3} \right) + \frac{x-2}{3} \cdot \frac{1}{\sqrt{9 - (x-2)^2}} \right] + C_1$$

$$18 \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$g \left(\theta + \frac{\sin 2\theta}{2} \right) + c$$

$$9 \left[\theta + 2 \sin \theta \cos \theta \right]$$

$$5) \int_0^{\frac{\pi}{4}} \cos 3x \sin^2 x \, dx$$

$$\frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 3x (1 - \cos 2x) \, dx$$

$$\frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos 3x - \cos 3x \cos 2x] \, dx$$

$$\frac{1}{2} \left[\frac{\sin 3x}{3} - \frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos(3x-2x) + \cos(3x+2x)] \, dx \right]$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} [\cos x + \cos 5x] \, dx$$

$$= \frac{1}{2} \left[\sin x + \frac{\sin 5x}{5} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{\sin 3x}{3} - \frac{1}{2} \left(\sin x + \frac{\sin 5x}{5} \right) \right]_0^{\frac{\pi}{4}}$$

$$\frac{1}{2} \left[\frac{\sin \frac{3\pi}{4}}{3} - \frac{1}{2} \left(\sin \frac{\pi}{4} + \frac{\sin 5\pi}{5} \right) \right]$$

$$\frac{1}{2} \left[\frac{\sin \frac{\pi}{4}}{3} - \frac{1}{2} \left(\frac{1}{\sqrt{2}} + -\frac{\sin \frac{\pi}{4}}{5} \right) \right]$$

$$\frac{1}{2} \left[\frac{\frac{1}{\sqrt{2}}}{3} - \frac{1}{2} \left(\frac{1}{\sqrt{2}} - \frac{\frac{1}{\sqrt{2}}}{5} \right) \right]$$

$$6) \int \cos 2x \ln(\tan x) dx$$

$$u = \ln(\tan x)$$

$$du = \frac{\sec^2 x}{\tan x} dx$$

$$= \frac{\frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x}}$$

$$= \frac{1}{\sin x \cos x} dx$$

$$dv = \cos 2x dx$$

$$v = \frac{\sin 2x}{2}$$

$$\frac{1}{2} \ln(\tan x) \sin 2x - \frac{1}{2} \int \sin 2x \frac{1}{\sin x \cos x} dx$$

$$= \frac{1}{2} \int \frac{2 \sin x \cos x}{\sin x \cos x} \frac{1}{\sin x \cos x} dx$$

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$$\int 1 dx$$

$$x + C$$

$$\Rightarrow \frac{1}{2} \ln(\tan x) \sin 2x - x + C$$

$$7) \int_e^{\infty} \frac{dx}{x(\ln x)^5}$$

(8)

$$I_1 = \int \frac{dx}{x(\ln x)^5}$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\int \frac{1}{u^5} du = \int u^{-5} du = \frac{u^{-4}}{-4} + C$$

$$= -\frac{1}{4} \frac{1}{u^4} + C = -\frac{1}{4} \frac{1}{(\ln x)^4} + C$$

$$\lim_{t \rightarrow \infty} \int_e^t \frac{dx}{x(\ln x)^5}$$

$$\lim_{t \rightarrow \infty} \left[-\frac{1}{4} \frac{1}{(\ln x)^4} \right]_e^t$$

$$= -\frac{1}{4} \lim_{t \rightarrow \infty} \left[\frac{1}{(\ln t)^4} - \frac{1}{(1)^4} \right]$$

$$= -\frac{1}{4} \left[\frac{1}{\infty} - 1 \right] = \frac{1}{4}$$

Convergent

Answer all questions. Calculators and mobile phones are NOT allowed.

1. (10pts each.) Find the following limits.

(a) $\lim_{x \rightarrow 0^+} (\cot x)^x.$

(b) $\lim_{x \rightarrow 0^+} (e^x - 1) \ln x.$

2. (10pts.) Find the volume of the solid obtained by revolving the region bounded by the graphs of the equations $y = \sin x$, $y = \cos x$ and $x = 0$, about the x -axis.

3. (10pts each.) Find the following integrals.

(a) $\int \ln(x^2 + 6x + 13) dx$

(b) $\int (\sinh^2 x) (\coth^5 x) dx$

(c) $\int \frac{3x^2 + x + 4}{x^3 + x} dx$

(d) $\int \frac{dx}{x - 4 + \sqrt{x - 2}}.$

4. (10pts.) Suppose that $f(1) = 3$, $f(4) = 4$, $f'(1) = -2$, $f'(4) = 3$, and f'' is continuous. Find the value of $\int_1^4 x f''(x) dx$.

5. (10pts each.) Determine whether each of the following integrals is convergent or divergent. Evaluate those that are convergent.

(a) $\int_2^\infty \frac{dx}{(4 + x^2)^2}.$

(b) $\int_0^4 \frac{dx}{(x - 1)^{4/3}}.$

1. (a) $\lim_{x \rightarrow 0^+} (\cot x)^x = e^{\lim_{x \rightarrow 0^+} x \ln \cot x}$.
 $\lim_{x \rightarrow 0^+} x \ln \cot x = \lim_{x \rightarrow 0^+} \frac{\ln \cot x}{\frac{1}{x}} = -\lim_{x \rightarrow 0^+} \frac{-\csc^2 x}{\frac{-1}{x^2}} = \lim_{x \rightarrow 0^+} \frac{x^2}{\sin^2 x \cot x} =$
 $\lim_{x \rightarrow 0^+} \frac{x^2}{\sin x \cos x} = 2 \lim_{x \rightarrow 0^+} \frac{x^2}{\sin 2x} = 0.$
 $\lim_{x \rightarrow 0^+} (\cot x)^x = 1.$
- (b) $\lim_{x \rightarrow 0^+} (e^x - 1) \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{(e^x - 1)^{-1}} = -\lim_{x \rightarrow 0^+} \frac{(e^x - 1)^2}{x e^x} = -2 \lim_{x \rightarrow 0^+} \frac{e^x (e^x - 1)}{x e^x + e^x}$
 $= -2 \lim_{x \rightarrow 0^+} \frac{e^x - 1}{x + 1} = 0.$
2. The curves intersect when $\sin x = \cos x$, so $x = \pi/4$. In our region $\cos x \geq \sin x$, so the volume is $\pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x) dx = \pi \int_0^{\pi/4} \cos 2x dx = \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} = \frac{\pi}{2}.$
3. (a) $\int \ln(x^2 + 6x + 13) dx = x \ln(x^2 + 6x + 13) - \int \frac{x(2x + 6)}{x^2 + 6x + 13} dx$
 $= x \ln(x^2 + 6x + 13) - \int \left(2 - \frac{6x + 26}{x^2 + 6x + 13} \right) dx$
 $= x \ln(x^2 + 6x + 13) - 2x + 2 \int \frac{3x + 13}{(x + 3)^2 + 4} dx =$
 $x \ln(x^2 + 6x + 13) - 2x + 2 \int \frac{3u + 4}{u^2 + 4} du, (u = x + 3)$
 $= -2x + (x + 3) \ln(x^2 + 6x + 13) + 4 \arctan\left(\frac{x + 3}{2}\right) + C.$
- (b) $\int (\sinh^2 x) (\cosh^5 x) dx = \int (\sinh^{-3} x) (\cosh^5 x) dx$
 $= \int (\sinh^{-3} x) (1 + \sinh^2 x)^2 d(\sinh x) = \int u^{-3} (1 + u^2)^2 du, (u = \sinh x)$
 $= 2 \ln u + \frac{u^2}{2} - \frac{1}{2u^2} = 2 \ln \sinh x + \frac{\sinh^2 x}{2} - \frac{1}{2 \sinh^2 x} + C.$
- (c) $\frac{3x^2 + x + 4}{x^3 + x} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$, which gives $3x^2 + x + 4 = Ax^2 + A + Bx^2 + Cx$ and $A = 4, B = -1, C = 1$. Thus
 $\int \frac{3x^2 + x + 4}{x^3 + x} dx = \int \left(\frac{4}{x} - \frac{x}{x^2 + 1} + \frac{1}{x^2 + 1} \right) dx = 4 \ln|x| - \frac{1}{2} \ln(x^2 + 1) + \tan^{-1} x + C_1.$
- (d) Put $x - 2 = t^2$. Then $dx = 2t dt$ and $\int \frac{dx}{x - 4 + \sqrt{x - 2}} = \int \frac{2t dt}{t + t^2 - 2} =$
 $= \int \frac{2t dt}{(t - 1)(t + 2)} = \int \left(\frac{2}{3} \frac{1}{t - 1} + \frac{4}{3} \frac{1}{t + 2} \right) dt = \frac{2}{3} \ln|\sqrt{x - 2} - 1| + \frac{4}{3} \ln(2 + \sqrt{x - 1}) + C.$
4. By parts: $\int_1^4 x f''(x) dx = 4f'(4) - f'(1) - \int_1^4 f'(x) dx = 4f'(4) - f'(1) - f(4) + f(1) =$
 $12 + 2 - 4 + 3 = 13.$
5. (a) Put $x = 2 \tan \theta$. Then $dx = 2 \sec^2 \theta d\theta$ and $\int \frac{dx}{(4 + x^2)^2} = \frac{1}{8} \int \cos^2 \theta d\theta$
 $= \frac{1}{16} \int (1 + \cos 2\theta) d\theta$. Thus $\int_2^\infty \frac{dx}{(4 + x^2)^2} = \lim_{t \rightarrow \infty} \int_2^t \frac{dx}{(4 + x^2)^2}$
 $= \frac{1}{16} \lim_{t \rightarrow \infty} (\theta + \frac{1}{2} \sin 2\theta) \Big|_{\pi/4}^{\tan^{-1} \frac{t}{2}}$
 $= \frac{1}{16} \lim_{t \rightarrow \infty} (\tan^{-1} \frac{t}{2} + \frac{1}{2} \sin(2 \tan^{-1}(t/2))) - \frac{\pi}{4} - \frac{1}{2}$
 $= \frac{1}{16} (\pi/2 - \pi/4 + 0 - 1/2) = \frac{1}{64} (\pi - 2).$
- (b) The integrand is discontinuous at $x = 1$ and $\int_1^4 \frac{dx}{(x - 1)^{4/3}} = \lim_{t \rightarrow 1^+} \int_t^4 \frac{dx}{(x - 1)^{4/3}} =$
 $\lim_{t \rightarrow 1^+} -3/\sqrt[3]{x - 1} \Big|_t^4 = \lim_{t \rightarrow 1^+} (-\frac{3}{\sqrt[3]{3}} + \frac{3}{\sqrt[3]{t - 1}}) = \infty$, so the integral diverges.

SOLUTIONS

Q1. The limit is indeterminate of the form $\frac{0}{0}$. Using l'Hospital's rule we obtain

$$\lim_{x \rightarrow 0^+} \frac{\arcsin \sqrt{x}}{1 - \ln(e - x)} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}}}{\frac{1}{e-x}} = \lim_{x \rightarrow 0^+} \frac{e-x}{2\sqrt{x}\sqrt{1-x}} = \infty.$$

Q2. Integrating by parts with $u = x^2$ and $dv = 4^x dx$ gives

$$I = \int x^2 4^x dx = \frac{1}{\ln 4} x^2 4^x - \frac{2}{\ln 4} \int x 4^x dx.$$

Integrating by parts again with $u = x$ and $dv = 4^x dx$ we obtain

$$I = \frac{1}{\ln 4} x^2 4^x - \frac{2}{(\ln 4)^2} x 4^x + \frac{2}{(\ln 4)^2} \int 4^x dx = \frac{1}{\ln 4} x^2 4^x - \frac{2}{(\ln 4)^2} x 4^x + \frac{2}{(\ln 4)^3} 4^x + C.$$

Q3. Completing the square under the root we have $x^2 - 4x + 3 = (x - 2)^2 - 1$. Let $\theta = \operatorname{arcsec}(x - 2)$, thus $x - 2 = \sec \theta$. Then

$$\begin{aligned} \int \frac{x dx}{\sqrt{x^2 - 4x + 3}} &= \int \frac{x dx}{\sqrt{(x-2)^2 - 1}} = \int \frac{(\sec \theta + 2) \sec \theta \tan \theta}{\tan \theta} d\theta \\ &= \int (\sec^2 \theta + 2 \sec \theta) d\theta = \tan \theta + 2 \ln |\sec \theta + \tan \theta| + C \\ &= \sqrt{x^2 - 4x + 3} + 2 \ln |x - 2 + \sqrt{x^2 - 4x + 3}| + C. \end{aligned}$$

Q4. Using the Weierstrass substitution $t = \tan(x/2)$ we obtain

$$I = \int \frac{dx}{2 \sin x - \cos x} = \int \frac{\frac{2}{1+t^2}}{2 \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} dt = \int \frac{2}{t^2 + 4t - 1} dt = \int \frac{2}{(t+2)^2 - 5} dt.$$

Now let $u = t + 2$ to obtain

$$I = 2 \int \frac{du}{u^2 - 5} = \frac{1}{\sqrt{5}} \ln \left| \frac{u - \sqrt{5}}{u + \sqrt{5}} \right| + C = \frac{1}{\sqrt{5}} \ln \left| \frac{\tan(x/2) + 2 - \sqrt{5}}{\tan(x/2) + 2 + \sqrt{5}} \right| + C.$$

Q5. Let $u = \sin x$. Then

$$\begin{aligned} \int \sin^5 x \cos^3 x dx &= \int \sin^5 x (1 - \sin^2 x) \cos x dx = \int u^5 (1 - u^2) du = \int (u^5 - u^7) du \\ &= \frac{1}{6} u^6 - \frac{1}{8} u^8 + C = \frac{1}{6} \sin^6 x - \frac{1}{8} \sin^8 x + C. \end{aligned}$$

Q6. Let $u = \ln x$. Then

$$\int_e^\infty \frac{dx}{x (\ln x)^3} = \lim_{t \rightarrow \infty} \int_e^t \frac{dx}{x (\ln x)^3} = \lim_{t \rightarrow \infty} \int_1^{\ln t} u^{-3} du = \lim_{t \rightarrow \infty} \left[\frac{u^{-2}}{-2} \right]_1^{\ln t} = -\frac{1}{2} \lim_{t \rightarrow \infty} \left[\frac{1}{(\ln t)^2} - 1 \right] = \frac{1}{2}.$$